

Inference for Spatial Dynamic Panel Model with Different Spatial Dependence Characterizations

Hui Wang^{*1}, Cheng Qian^{†2}, and Clifford Lam^{‡2}

¹Department of Financial Engineering, School of Finance , Central University of Finance and Economics, China

²Department of Statistics, London School of Economics and Political Science, UK

Abstract. This paper proposes a generalized spatial panel model in the sense that a spatial weights matrix is constructed through incorporating different spatial dependence characterizations using a linear combination of them, while the errors of the model can be both serially and cross correlated. The consistency and asymptotic normality of our estimators are established when n , the dimension, and T , the sample size, are both large. Based on the asymptotic results, we propose a Wald test to test both the contemporaneous and time-lagged spatial correlations. Furthermore, a diagnostic test is proposed to test if the fitted residuals, having dimension n which can be growing with T , behave like a white noise. Our method is demonstrated thoroughly through simulations and a stock return analysis.

Key words: Quasi-maximum likelihood estimation, different spatial dependence, spatial dynamic panel data, spatial correlation tests, high dimensional diagnostic test.

^{*}Hui Wang is Professor, Department of Financial Engineering, School of Finance, Central University of Finance and Economics. Email: xiaohuipk@163.com

[†]Cheng Qian is PhD student, Department of Statistics, London School of Economics. Email: C.Qian2@lse.ac.uk

[‡]Clifford Lam is Associate Professor, Department of Statistics, London School of Economics. Email: C.Lam2@lse.ac.uk

1 Introduction

The study of spatial dynamic panel data models has seen a lot of theoretical advancements in recent years. The generalized method of moments (Kapoor et al., 2007, Lee and Yu, 2014) and the quasi-maximum likelihood estimation (Lee and Yu, 2010, Yu et al., 2008) are two approaches that are extensively developed.

However, the papers above assume that the spatial weights matrix in their models is known *a priori*. Applied researchers often based on their subject knowledge in pre-setting such a matrix, which can be inaccurate and subjective at times. Recognizing the importance of correct specification of a spatial weights matrix in the accuracy of estimated model parameters, a surge of papers have focused on estimating a spatial weights matrix using data. These include the use of Lasso or the adaptive Lasso to perform sparse estimation of the spatial weight matrix in a spatial lag panel model (Ahrens and Bhattacharjee, 2015, Lam and Souza, 2014, 2015), the use of models with several spatial weights matrices (Arnold et al., 2013, Badinger and Egger, 2011, Lee and Liu, 2010, LeSage and Pace, 2008), or both (Lam and Souza, 2016).

Observing that in many applications, researchers often has much more than one way to pre-specify a spatial weight matrix, we consider a spatial dynamic panel model allowing for both contemporaneous and time-lagged spatial correlations, while the errors can be both serially and cross correlated. The two types of spatial correlations are estimated through two different linear combinations of pre-specified spatial weights matrices. Our proposed model can also include time series covariates as input. With the use of instrumental variables and profile least square technique, Lam and Qian (2017) has proved some useful asymptotic results for a similar model. Since we are using quasi-maximum likelihood estimation, we do not need the existence of instrumental variables for the estimation of our model.

Our paper has three main contributions. Firstly, we extend the quasi-maximum likelihood estimation technique in estimating a spatial dynamic panel data model which allows for different spatial weight matrices representing the contemporaneous and time-lagged spatial dependence. Individual fixed effects are absorbed into our covariates (see Section 2), so that without any time fixed effects, our model and estimation technique are generalization of those in Lee and Yu (2010). Secondly, we provide a unified spatial test statistic for testing the presence of either

contemporaneous or time-lagged spatial correlation, or both. Finally, we provide a diagnostic test for testing if the fitted residuals are white noise. This diagnostic test allows for the dimension n of the fitted residual vector to be growing with the sample size T . To the best of our knowledge, in spatial econometrics context, this is the first time a high dimensional diagnostic test is available for model checking, which is a fundamental part of data analysis. Our test does not only test the whiteness of the fitted residuals themselves, but also can conclude if the underlying error series are truly white noise.

The rest of the paper is organized as follows. Section 2 presents our model, setting up notations and present a sufficient condition for stationarity. Section 3 describes several areas where our model can be useful and how it is a generalization of commonly used models. The quasi-maximum likelihood estimation, together with necessary assumptions and asymptotic results for our estimators are presented in Section 4. Section 5 presents the Wald statistic and the corresponding asymptotic theory for testing different spatial correlations, while Section 6 provides the test statistic and presents the asymptotic theory for the high dimensional diagnostic test for white noise in our model. Section 7 presents our simulation results as well as a stock return analysis. All technical proofs are relegated to Section 8, with Appendix A detailing how we find the derivatives of the log-likelihood function of our model.

2 The model

Consider, for $t = 1, \dots, T$, the model

$$Y_{nt} = \left(\sum_{i=1}^r \alpha_{0i} W_{ni} \right) Y_{nt} + \left(\sum_{i=1}^r \gamma_{0i} W_{ni} \right) Y_{n,t-1} + \phi_0 Y_{n,t-1} + X_{nt} \beta_0 + \varepsilon_{nt}, \quad (2.1)$$

where $Y_{nt} = (y_{1t}, \dots, y_{nt})'$ and $\varepsilon_{nt} = (e_{1t}, \dots, e_{nt})'$ are n dimensional vectors, e_{it} is an error series with zero mean and variance σ_0^2 , $\{W_{ni}, i = 1, \dots, r\}$ are $n \times n$ predetermined spatial weights matrices. Finally, X_{nt} is an $n \times \kappa_x$ matrix of non-stochastic regressors.

Model (2.1) is different from a traditional spatial dynamic panel model in the following two aspects. First, we consider a general spatial dependence, which can be decomposed as a linear combination of pre-specified “spatial dependence”, called spatial weights matrices. We then estimate this linear combination through data, so that information from traditional expert knowledge

on spatial relationship can be merged with microstructure of data to enhance the accuracy of the resulting spatial dependence structure.

Secondly, to reduce the parameter dimension, we model the fixed effects μ_{0i} by $\mu_{0i} = \mu_0 + Z_i' \eta$, where μ_0 is the intercept shared by all the components and Z_i is a p -dimensional component specific covariate vector, where p is a fixed integer. We then embed this into the covariate matrix X_{nT} , so that model (2.1) does include the fixed effects for all the components in Y_{nt} .

Denote $\alpha = (\alpha_1, \dots, \alpha_r)' \in \Lambda$, with true value $\alpha_0 = (\alpha_{10}, \dots, \alpha_{r0})'$, and $\gamma = (\gamma_1, \dots, \gamma_r)' \in \Gamma$, with true value $\gamma_0 = (\gamma_{10}, \dots, \gamma_{r0})'$. For convenience of our presentation, we also define $\varphi = (\gamma', \phi)'$, $\delta = (\gamma', \phi, \beta')'$, $\xi = (\alpha', \delta')'$ and $\theta = (\xi', \sigma^2)'$, with similar definitions for their respective true values. Define

$$H_n(\alpha) = I_n - \sum_{i=1}^r \alpha_i W_{ni}, \quad \Upsilon(\varphi) = \sum_{i=1}^r \gamma_i W_{ni} + \phi I_n, \quad A_n(\alpha, \varphi) = H_n^{-1}(\alpha) \Upsilon_n(\varphi).$$

Then, Y_{nt} can be rewritten as

$$Y_{nt} = A_n Y_{n,t-1} + H_n^{-1} X_{nt} \beta_0 + H_n^{-1} \varepsilon_{nt}, \quad (2.2)$$

where $H_n = H_n(\alpha_0)$ and $A_n = A_n(\alpha_0, \varphi_0)$.

From (2.2), model (2.1) can be embedded into the framework of a vector autoregressive (VAR) model. By Corollary (5.6.16) of Horn and Johnson (2012), a sufficient condition for the stationarity of model (2.1) is $\|A_n\| < 1$, where the matrix norm can be either the L_1 or L_∞ norm, defined by $\|M\|_1 = \max_j \sum_i |m_{ij}|$ and $\|M\|_\infty = \max_i \sum_j |m_{ij}|$ respectively. If the W_{ni} 's are row-standardized, in the sense that the sum of the absolute value of the elements in each row of W_{ni} equals to 1 (see for example, LeSage and Pace (2009) for the use of such row-standardization), then we have $\|\sum_{i=1}^r \alpha_{0i} W_{ni}\|_\infty \leq \sum_{i=1}^r \alpha_{0i}$. By Lemma 2.3.3 of Golub and van Loan (1996), we obtain that

$$\|H_n^{-1}\|_\infty \leq \frac{1}{1 - \sum_{i=1}^r \alpha_{0i}}.$$

Note that

$$\|A_n\|_\infty \leq \|H_n^{-1}\|_\infty \|\Upsilon_n\|_\infty \leq \frac{\sum_{i=1}^r |\gamma_{0i}| + |\phi_0|}{1 - \sum_{i=1}^r |\alpha_{0i}|}.$$

Therefore, $\sum_{i=1}^r |\alpha_{0i}| + \sum_{i=1}^r |\gamma_{0i}| + |\phi_0| < 1$ is a sufficient condition for $\|A_n\|_\infty < 1$, and hence stationarity of model (2.2), which is assumed throughout the paper.

3 Some application examples

The proposed model can be very useful in a variety of economic and social setups. In the following, we give three examples.

1. **Economics.** Economic activities are often concentrated geographically. The development of policies for strengthening economic growth, for example, is therefore dependent on the understanding of the clustering of such activities spatially. Spatial econometric models, through the specification of a spatial weights matrix, is designed to account for spatial interactions among observed units, allowing for the effects of exogenous variables.

An example in studying the growth of different European regions is Baumont et al. (2003). In it, different distance-based spatial weights matrices W are explored, and plots of log per-capita GDP for different European countries (Y) against the spatial-lagged ones (WY) consistently reveal two clusters of regions in Europe, at the same time showing a consistent positive slope, called the spatial correlation. In essence, it suggests the use of a model of the form

$$Y = \rho WY + X\beta + \epsilon,$$

where ρ is the spatial correlation, and X is a matrix of exogenous covariates, can be useful. Such a spatial lag model is discussed in details in Anselin (2010).

While the model can be useful, it is generally agreed that the specification of W has to be given extreme care, since different W can give substantially different results. Abreu et al. (2005) suggests that no matter a contiguity matrix (where neighboring regions are coded 1 and others as 0) or distance-based matrices are used, they should arise from underlying theoretical considerations, on top of being exogenous from the variables that are being studied. This creates difficulties in practice since even for distance based matrices, using d^{-1} or d^{-2} can be potentially different. Indeed, Baumont et al. (2003) explored different distance-based spatial weights matrices before coming to their conclusion of regional polarization.

The model (2.1) we proposed can handle this situation nicely by considering a linear combination of some sensible spatial weights matrices, and let the “best” spatial weights matrix to be estimated from the data as the “best” linear combination. Lee and Liu (2010) also con-

sider a model with a linear combination of spatial weights matrices, and called this higher order spatial lags, with generalized method of moments proposed to estimate the model. Our model can be considered a generalization of theirs since we also include a time and spatial-lagged term $(\phi_0 + \sum_{i=1}^r \gamma_{0i} W_{ni}) Y_{n,t-1}$. In the case of regional growth by country for instance, a time-lagged shock from neighboring regions of a country can still carry an effect to the current GDP growth of the particular country. Such a shock can be well estimated by our model with such a time and spatial-lagged term.

2. **Social Network.** Social network represents general relationships among individual units. A prominent example is the Twitter or Facebook network, where an individual is linked to another person if he or she is “following” or “friend” of another. In Bramoullé et al. (2009), the level of a student’s recreational activity y_i is modeled as

$$y_i = \alpha + \beta \frac{\sum_{j \in P_i} y_j}{n_i} + \gamma x_i + \delta \frac{\sum_{j \in P_i} x_j}{n_i} + \epsilon_i, \quad E(\epsilon_i | \mathbf{x}) = 0,$$

where P_i is the set of friends for individual i with $n_i = |P_i|$, and x_j is the parents’ income for the j th individual. The model clearly includes the mean of the friends’ level as an endogenous factor for the recreational level of individual i , which can be considered a peer effect. His or her parents’ income, as well as the mean parents’ income of all friends, are also intuitively important factors that are included in the model above. This particular social network model can actually be captured by our model with $T = r = 1$, $\alpha_{01} = \beta$, $\gamma_{01} = \phi_0 = 0$, $W_{n1} = (1/n_i \cdot 1_{\{j \in P_i\}})$ and

$$X_{n1} = \begin{pmatrix} 1 & x_1 & x_1/n_1 \cdot 1_{\{1 \in P_1\}} & \cdots & x_n/n_1 \cdot 1_{\{n \in P_1\}} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & x_1/n_n \cdot 1_{\{1 \in P_n\}} & \cdots & x_n/n_n \cdot 1_{\{n \in P_n\}} \end{pmatrix}, \quad \beta_0 = (\alpha, \gamma, \delta \mathbf{1}'_n)',$$

where $\mathbf{1}_n$ is a column vector of n ones, and 1_A is 1 under the event A and 0 otherwise. If $n_i < \infty$, then each non-zero entry in the i th row of X_{n1} (apart from the constant 1) can be the entries in the covariate vector Z_i in our model (with some entries possibly 0 if the n_i ’s are not all equal), with $p = \max(n_i) + 1$ and $\eta = (\gamma, \delta \mathbf{1}'_{\max(n_i)})'$. Clearly, our model is more general than that introduced in Bramoullé et al. (2009), with the possibility of adding time-lagged peer effects.

Our model is also a generalization of the network vector autoregression proposed by Zhu et al. (2016). Compared to the model (2.1) in their paper, we have the spatial interaction term $\sum_{i=1}^r \alpha_{0i} W_{ni}$, which can be important to capture the endogenous peer effects in a social network. At the same time, setting $W_{n1} = (a_{ij}/n_i)$, $\gamma_{01} = \beta_1$ and $\phi_0 = \beta_2$ in our model, we have the autoregressive term in model (2.1) of Zhu et al. (2016). The analysis of our paper is also significantly different from that in Zhu et al. (2016) because of the endogenous peer effects term.

3. **Finance.** Spatial dependence can be used in different areas of finance for interpretation/forecasting purposes. One important interest lies in the spatial dependence among different financial markets. This is closely related to the contagion of financial markets, and hence how world market shocks are propagated to one another. In Fernandez (2011), a spatial Capital Asset Pricing Model is used for modeling asset returns in several emerging markets, where such a model is in parallel to the spatial lag model, giving an extra risk term coming from a weighted average of neighboring firms on top of the usual risk premium from the market portfolio. In doing so, four different spatial weights matrices are calculated based on four different financial indicators. They are market capitalization (relative to firm size), the market-to-book, the dividend yield and the debt maturity ratios. Subsequent analyses use these spatial weights matrices separately for drawing their conclusions. Our model, however, can accommodate these different spatial weights matrix specifications by finding a linear combination of them that is more adapted to the data. This can also enhance any conclusions that can be drawn only separately from using different spatial weights matrices individually.

Indeed, Arnold et al. (2013) has used a linear combination of spatial weights matrices when using a spatial lag model without exogenous covariates to model stock returns. In their notations, they model the vector of stock returns y_t by

$$y_t = \rho_g W_g y_t + \rho_b W_b y_t + \rho_l W_l y_t + \epsilon_t,$$

where W_g , W_b and W_l are spatial weights matrices such that $W_g y_t$ represents the weighted market return of the same day, $W_b y_t$ the weighted market return of the respective industrial branches, and finally $W_l y_t$ the weighted local market return of the respective countries.

This simple model already gives very good Value-at-Risk forecasts. The model we proposed certainly generalizes theirs with the possibility of adding important exogenous covariates, as well as time-lagged spatial dependence.

4 The quasi-maximum likelihood estimators

The total number of parameters of model (2.1) is $2r + \kappa_x + 2$. The log-likelihood function of model (2.1) is

$$\log L_{nT}(\theta) = -\frac{nT}{2} \log 2\pi - \frac{nT}{2} \log \sigma^2 + T \log |H_n(\alpha)| - \frac{1}{2\sigma^2} \sum_{t=1}^T \varepsilon'_{nT}(\xi) \varepsilon_{nt}(\xi), \quad (4.1)$$

where

$$\varepsilon_{nt}(\xi) = H_n(\alpha)Y_{nt} - \Upsilon_n(\varphi)Y_{n,t-1} - X_{nt}\beta. \quad (4.2)$$

The quasi-maximum likelihood estimator (QMLE) $\hat{\theta}_{nT}$ is defined as

$$\hat{\theta}_{nT} = \arg \min_{\theta \in \Theta} l_{nT}(\theta),$$

where Θ is the parameter space and

$$l_{nT}(\theta) = -\frac{1}{nT} \log L_{nT}(\theta). \quad (4.3)$$

To obtain the asymptotic results of QMLE $\hat{\theta}_{nT}$, we need the following assumptions.

- A1.** The spatial weights matrix W_{ni} is a constant, with zero diagonal elements for each $i = 1, \dots, r$, $j = 1, \dots, n$.
- A2.** The error series $\{e_{it}\}$ are independent across i and are martingale differences across t , with $Ee_{it} = Ee_{it}^3 = 0$ and $Ee_{it}^2 = \sigma_0^2$. Furthermore,
 - (i) there exists a constant $\tau > 0$ such that $E|e_{it}|^{4+\tau} < \infty$;
 - (ii) the quantity $\sum_{h=0}^{\infty} \Gamma(h)$ is finite, with $\Gamma(h) = \text{cov}(e_{it}^2, e_{i,t+h}^2)$.
- A3.** The matrix $H_n(\alpha)$ is invertible for all $\alpha \in \Lambda$, where Λ is compact and α_0 is in the interior of Λ .

- A4.** The elements of X_{nt} are non-stochastic and bounded uniformly in n and t . Furthermore, $\lim_{t \rightarrow \infty} \frac{1}{nT} \sum_{t=1}^T X'_{nt} X_{nt}$ exists and is nonsingular.
- A5.** The quantities $\|W_{ni}\|_1$ and $\|W_{ni}\|_\infty$ are both bounded uniformly in n for $i = 1, \dots, r$. The same goes for $\|H_n^{-1}(\alpha)\|_1$, $\|H_n^{-1}(\alpha)\|_\infty$.
- A6.** The quantities $\|\sum_{h=1}^\infty \text{abs}(A_n^h)\|_1$ or $\|\sum_{h=1}^\infty \text{abs}(A_n^h)\|_\infty$ are bounded, where $\text{abs}(A)$ is the matrix of absolute values of the entries in A .
- A7.** The dimension n is a nondecreasing function of T as T goes to infinity.
- A8.** The matrix $\lim_{T \rightarrow \infty} E\mathcal{J}_{nT}$ is nonsingular, where $\mathcal{J}_{nT} = \frac{1}{nT} \sum_{t=1}^T (\tilde{O}_{nT}, O_{nt})'(\tilde{O}_{nT}, O_{nt})$ with $\tilde{O}_{nT}$ and O_{nt} defined by (A.4) and (8.1) respectively.

Remark 1 Assumption A1 is standard for spatial econometric models. Assumption A2 allows for dependence of the e_{it} 's at different time points, which relaxes the usual independence and identical distribution assumptions for consistency. The requirement $Ec_{it}^3 = 0$ is a technical condition for a more explicit expression for the global identification of the parameters. Assumption A3 and A6 ensure that model (2.1) has a moving average representation, and compactness is a technical condition. Assumption A4 is for the convenience of proof and can be relaxed to X_{nt} being stochastic with some moments restrictions. Assumption A5 limits the spatial correlation to a manageable degree. Assumption A6 limits the dependence of y_t across time and space, and together with Assumption A2, A3 and A5 imply that model (2.1) has the moving average representation

$$Y_{nt} = \sum_{h=0}^{\infty} A_n^h H_n^{-1} (\varepsilon_{n,t-h} + X_{n,t-h} \beta_0) = \mathcal{E}_{nt} + \mathcal{X}_{nt}, \quad (4.4)$$

where $\mathcal{E}_{nt} = \sum_{h=0}^{\infty} A_n^h H_n^{-1} \varepsilon_{n,t-h}$ and $\mathcal{X}_{nt} = \sum_{h=0}^{\infty} A_n^h H_n^{-1} X_{n,t-h}$. Assumption A7 includes the fixed n and $n \rightarrow \infty$ scenarios. Finally, Assumption A8 is an identification condition for the parameters of model (2.1).

Theorem 1. *Suppose Assumption A1-A8 hold and $E\mathcal{A}_{nT}$ is nonsingular, then θ_0 is globally identified and $\hat{\theta}_{nT} \xrightarrow{P} \theta_0$.*

Theorem 2. *Suppose the conditions of Theorem 1 hold. Then, we have*

$$\sqrt{nT}(\hat{\theta}_{nT} - \theta_0) \xrightarrow{d} N(0, \Sigma^{-1}\Omega\Sigma^{-1}),$$

where

$$\Omega = \lim_{T \rightarrow \infty} nTE\left(\frac{\partial l_{nT}(\theta_0)}{\partial \theta} \frac{\partial l_{nT}(\theta_0)}{\partial \theta'}\right), \quad \Sigma = \lim_{T \rightarrow \infty} E\left(\frac{\partial^2 l_{nT}(\theta_0)}{\partial \theta \partial \theta'}\right).$$

Theorem 2 means that the QMLE is asymptotic normal with standard rate $\sqrt{nT}$ even if n is much larger than T . This is because we model the fix effects by some specific variables to reduce the dimension of the parameter to a fixed number.

5 The tests for spatial autocorrelation

It is very important to access whether the specification of the spatial autocorrelation is appropriate for the data. This section considers a unified framework for testing the linear relationship of the coefficients:

$$H_0 : a'\theta_0 = c \text{ against } H_1 : a'\theta_0 \neq c, \quad (5.1)$$

where a is a known vector of length $2r + 2 + \kappa_x$ and c is a known constant. With different values of a , (5.1) includes the following hypotheses.

- (i) $H_0^1 : \alpha_{0i} = 0, i = 1, \dots, r$. Under the null hypothesis, there is no contemporaneous spatial correlation.
- (ii) $H_0^2 : \gamma_{0i} = 0, i = 1, \dots, r$. Under the null hypothesis, there is no lag-1 spatial correlation.
- (iii) $H_0^3 : \alpha_{0i} = 0, \gamma_{0i} = 0, i = 1, \dots, r$. Under the null hypothesis, there is no spatial correlation at all.

Denote

$$\hat{\Sigma}_{nT} = \frac{\partial^2 l_{nT}(\hat{\theta}_{nT})}{\partial \phi \partial \phi'}, \quad \hat{\Omega}_{nT} = nT \frac{\partial l_{nT}(\hat{\theta}_{nT})}{\partial \theta} \frac{\partial l_{nT}(\hat{\theta}_{nT})}{\partial \theta'}.$$

We use a Wald test statistic for testing H_0 against H_1 in (5.1), which is defined as

$$\mathcal{W} = nT(a'\hat{\theta}_{nT} - c)'(a\hat{\Sigma}_{nT}^{-1}\hat{\Omega}_{nT}\hat{\Sigma}_{nT}^{-1}a')^{-1}(a'\hat{\theta}_{nT} - c).$$

Based on the asymptotic results in the last section, we have the following theorem.

Theorem 3. *Suppose the conditions of Theorem 2 hold. Then $\mathcal{W} \xrightarrow{d} \chi^2(1)$ under H_0 as $T \rightarrow \infty$.*

6 Diagnostic testing for the model

If the model is adequate, then the estimated errors should be a white noise. We adopt the maximum cross-correlations method proposed by Chang et al. (2017) for the estimated errors. Let

$$\hat{\rho}(k) = (\hat{\rho}_{ij}(k)) = \text{diag}(\hat{\Lambda}(0))^{-1/2} \hat{\Lambda}(k) \text{diag}(\hat{\Lambda}(0))^{-1/2}$$

be the sample autocorrelation for the estimated errors, where

$$\hat{\Lambda}(k) = \frac{1}{T} \sum_{t=1}^{T-k} \varepsilon_{n,t+k}(\hat{\xi}) \varepsilon'_{n,t}(\hat{\xi}). \quad (6.1)$$

Consider the hypothesis testing problem

$$H_0 : \{\varepsilon_{nt}\} \text{ is white noise.}$$

The test statistic is defined as

$$\mathcal{T}_{nT} = \max_{1 \leq k \leq K} \mathcal{T}_{n,k},$$

where $\mathcal{T}_{n,k} = \max_{1 \leq i,j \leq n} \sqrt{T} |\hat{\rho}_{ij}(k)|$ and K is a prescribed integer. We reject H_0 if $\mathcal{T}_{nT} > c_\eta$, where c_η is determined by

$$Pr(\mathcal{T}_{nT} > c_\eta | H_0) = \eta.$$

To get the critical value c_η , one needs to know the asymptotic distribution of $\mathcal{T}_{nT}$, which we unfortunately do not. Following the lines of Chang et al. (2017), we use the distribution of $\|\mathcal{G}\|_\infty$ to approximate the distribution of $\mathcal{T}_{nT}$, where $\mathcal{G}$ is a multivariate normal random vector with mean 0 and covariance Ψ_{nT} . We estimate Ψ_{nT} as follows. Let $\hat{\varepsilon}_{n,t} = \varepsilon_{n,t}(\hat{\xi})$, and

$$\begin{aligned} f_t &= (\text{vec}(\hat{\varepsilon}_{n,t+1} \hat{\varepsilon}'_{n,t}), \dots, \text{vec}(\hat{\varepsilon}_{n,t+K} \hat{\varepsilon}'_{n,t}))', \\ \hat{\mathcal{H}}(j) &= \begin{cases} \frac{1}{T-K} \sum_{t=j}^{T-K} f_t f'_{t-j}, & j \geq 0 \\ \frac{1}{T-K} \sum_{t=-j+1}^{T-K} f_{t+j} f'_t, & j < 0, \end{cases} \\ \mathcal{K}(x) &= \frac{25}{12\pi^2 x^2} \left[\frac{\sin(6\pi x/5)}{6\pi x/5} - \cos(6\pi x/5) \right]. \end{aligned}$$

Then, Ψ_{nT} is estimated by

$$\hat{\Psi}_{nT} = (I_K \otimes \hat{\mathcal{Z}}) \mathcal{Q}_{nT} (I_K \otimes \hat{\mathcal{Z}}),$$

where $\hat{\mathcal{Z}} = [\text{diag}(\hat{\Lambda}(0))]^{-1/2} \otimes [\text{diag}(\hat{\Lambda}(0))]^{-1/2}$ for $\hat{\Lambda}(0)$ defined in (6.1), b_T is a bandwidth diverging with T , and

$$\mathcal{Q}_{nT} = \sum_{j=-T+K+1}^{T-K-1} \mathcal{K}\left(\frac{j}{b_T}\right) \hat{\mathcal{H}}(j).$$

Define $\hat{c}_\eta$ by

$$Pr(\|\mathcal{G}\|_\infty > \hat{c}_\eta | \hat{\epsilon}_{n1}, \dots, \hat{\epsilon}_{nT}) = \eta,$$

where $\mathcal{G} \sim N(0, \hat{\Psi}_{nT})$. The value $\hat{c}_\eta$ can serve as the critical value for $\mathcal{T}_{nT}$ due to the following theorem, which needs 3 additional assumptions.

A9. There exist constants $c_1, c_2 > 0$ and $\tilde{r} \in (0, 2]$ such that $Pr(|e_{11}| > x) \leq C_1 \exp(-C_2 x^{\tilde{r}_1})$.

A10. The error ϵ_{nt} in model (2.1) is β -mixing with mixing coefficients satisfying $\tilde{\beta}_k \leq \exp(-c_3 k^{\tilde{r}_2})$ for some constants $c_3 > 0$ and $\tilde{r}_2 \in (0, 1]$.

A11. There exists a constant $c_4 > 0$ and $\zeta > 0$ such that

$$\frac{1}{c_4} \liminf_{q \rightarrow \infty} \inf_{m \leq 0} E\left(|\frac{1}{\sqrt{q}} \sum_{t=m+1}^{m+q} e_{i,t+k} e_{jt}|^{2+\zeta}\right) \leq \limsup_{q \rightarrow \infty} \sup_{m \leq 0} E\left(|\frac{1}{\sqrt{q}} \sum_{t=m+1}^{m+q} e_{i,t+k} e_{jt}|^{2+\zeta}\right) \leq c_4.$$

Theorem 4. Suppose Assumption A9-A11 and the conditions of Theorem 2 hold. If $b_n \sim T^\rho$ for $0 < \rho < \min\{1/6, \tilde{r}_2/(1 + \tilde{r}_2)\}$ and $\log(p) \leq CT^{\tilde{\rho}}$ for some constants C and $\tilde{\rho}$, then as $T \rightarrow \infty$,

$$Pr(\mathcal{T}_{nT} > \hat{c}_\eta | H_0) \rightarrow \eta.$$

7 Numerical Study

7.1 Performance of QMLE

We generate X_{nt} and ϵ_{nt} in model (2.1) as follows. Set $\kappa_x = 3$ (3 columns in X_{nT}) and we generate X_{nT} using $\text{vec}(X_{nT}) \sim N(\mathbf{0}, \Sigma_X)$, where

$$\Sigma_X = \begin{bmatrix} 2\mathbf{I}_n & 0.5\mathbf{I}_n & 0.5\mathbf{I}_n \\ 0.5\mathbf{I}_n & 2\mathbf{I}_n & 0.5\mathbf{I}_n \\ 0.5\mathbf{I}_n & 0.5\mathbf{I}_n & 2\mathbf{I}_n \end{bmatrix}.$$

The error ϵ_{nt} 's are independent of each other, each follows a multivariate normal distribution $N(\mathbf{0}, \mathbf{I}_n)$.

We set $r = 3$ for model (2.1). All elements in the parameters ϕ , α , γ and β are independently generated from the uniform distribution $U(0, 1)$ in each simulation run. To satisfy the sufficient condition for stationarity, every element in ϕ_0 , α_0 , γ_0 is divided by 1.2 times the absolute sum of all of them.

As $r = 3$, we need 3 specified spatial weights matrices W_{n1} , W_{n2} and W_{n3} . To simplify the simulation procedure so that each W_{ni} still has eigenvalues less than 1 in magnitude, which is a part of the stationarity conditions for model (2.1), we generate each W_{ni} with only the first three off-diagonals (both lower and upper) being non-zero. Another procedure where we produce sparse W_{ni} produces very similar results and are not presented in the paper.

We use the MATLAB function `fmincon` to evaluate the QMLE $\hat{\theta}_{nT}$ constrained for stationarity. Since the likelihood function is in general not convex, it is necessary to find a good initial value for the procedure. We apply the method in Lam and Qian (2017), which can provide an accurate least square type estimator if suitable instrumental-type variables exist. In our simulations, the X_{nt} 's are exogenous and can be the "instrumental variables". We also simulate X_{nT} to be correlated with ϵ_{nT} in general while still using the X_{nT} 's as the "instruments" needed in Lam and Qian (2017). It turns out that the initial values obtained are still good enough for QMLE $\hat{\theta}_{nT}$ to converge to reasonable values as long as the correlations between the elements in X_{nT} and ϵ_{nT} are not too strong, and hence the simulation results are very similar in the end.

We repeat our simulation 500 times. Figure 1 shows the boxplots of the averaged L_1 -error for $\hat{\theta}_{nT}$, which is $\|\hat{\theta}_{nT} - \theta_0\|_1/11$, under different combinations of (n, T) . It is clear that $\hat{\theta}$ is convergent to the true value as n or T becomes large.

To examine the asymptotic normality of $\hat{\theta}_{nT}$ in Theorem 2, we use the previous settings for 500 simulation runs. Using the results in Theorem 2, we calculate $\hat{\Sigma} = E(\partial^2 l_{nT}(\hat{\theta}_{nT})/\partial\theta\partial\theta')$ as an estimator for Σ . We choose $\hat{\sigma}^2$ from $\hat{\theta}_{nT}$ and standardize it using the true value σ_0^2 and the corresponding entry in the estimated covariance matrix $\hat{\Sigma}$ when $(n, T) = (100, 100)$. The histogram and normal probability plots in Figure 2 show that our standardized estimator follows a standard normal distribution. Other combinations of (n, T) produces similar results and are not shown here.

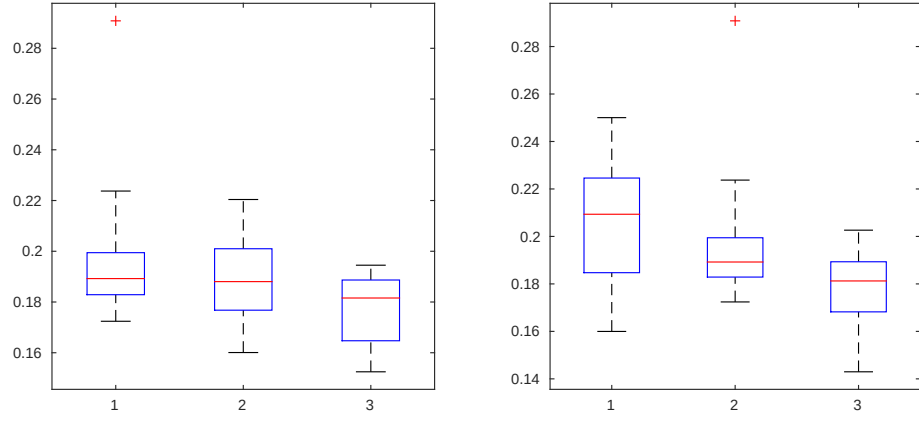


Figure 1: Boxplots of $\|\hat{\theta}_{nT} - \theta_0\|_1/11$. Left panel, from left to right: $n = 50, 100, 150, T = 100$. Right panel, left to right: $T = 50, 100, 150, n = 100$.

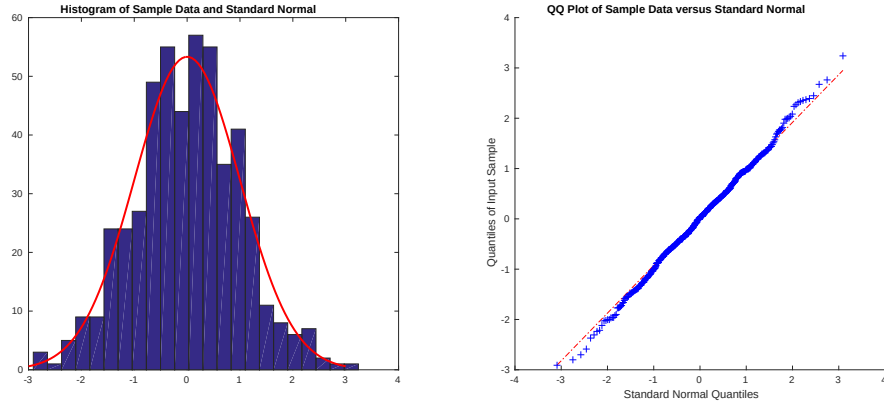


Figure 2: Histograms (left) and normal probability plot (right) for standardized $\hat{\sigma}^2$. Standardization used the estimated asymptotic covariance matrix derived in Theorem 2.

7.2 Performance of spatial and diagnostic tests

We use the settings in the previous section except for the values of α_0 and γ_0 , which are set according to three different scenarios:

1. Both contemporaneous and lag-1 spatial correlations are present, $\alpha_0 \neq 0, \gamma_0 \neq 0$;
2. Only lag-1 spatial correlation presents, $\alpha_0 = 0, \gamma_0 \neq 0$;
3. Only contemporaneous spatial correlation presents, $\alpha_0 \neq 0, \gamma_0 = 0$.

Under scenario 1, all null hypotheses in Section 5 should be rejected, while only H_0^2 and H_0^3 should be rejected under scenario 2. Under scenario 3, only H_0^1 and H_0^3 should be rejected. Table 1 shows the proportion of times H_0^1 , H_0^2 and H_0^3 are rejected in 500 simulation runs under the three different scenarios above. Significance level of the tests are set at 5%. It is clear that in many combinations of (n, T) , our spatial test is doing very well, rejecting the hypothesis that should not be rejected below 5% of times, and at the same time rejecting those hypotheses that should be rejected usually at or above 95% of times.

For the diagnostic test in Section 6, since we generate our data using white noise as errors, ideally we should not reject our null hypothesis of ε_{nT} being a white noise. With significance level set at 5% and $K = 10$ lags considered in the test, Table 2 shows the results. As n or T gets larger, the proportion of not rejecting the null hypothesis is getting closer to 95%.

7.3 Power of the diagnostic test

To see the power of our proposed diagnostic test, we consider generating ε_{nt} as follows. Let ε_{nt} follows a vector autoregressive process of order one, defined by $\varepsilon_{nt} = a\mathbf{I}_n\varepsilon_{nt-1} + z_{nt}$, where the z_{nt} 's are independent of each other and all elements in z_{nt} are generated from the t_8 distribution. All other settings are the same as in Section 7.1. When $a = 0$, ε_{nt} is a white noise, and so the rejection rate of our diagnostic test should ideally be at the nominal level. As a increases, rejection rate should increase for our test, and eventually approach one.

Figure 3 shows the power curves for different combinations of n and T . As T increases with n fixed, the power of the diagnostic test generally increases, while it decreases in general when

	$\alpha \neq 0$ and $\gamma \neq 0$								
	H_0^1			H_0^2			H_0^3		
	$T = 50$	$T = 100$	$T = 150$	$T = 50$	$T = 100$	$T = 150$	$T = 50$	$T = 100$	$T = 150$
$n = 25$	93%	96%	97%	90%	92%	99%	93%	95%	99%
$n = 50$	91%	92%	99%	95%	98%	99%	90%	95%	99%
$n = 75$	95%	99%	99%	91%	99%	99%	95%	98%	99%
	$\alpha = 0$ and $\gamma \neq 0$								
	H_0^1			H_0^2			H_0^3		
	$T = 50$	$T = 100$	$T = 150$	$T = 50$	$T = 100$	$T = 150$	$T = 50$	$T = 100$	$T = 150$
$n = 25$	1%	5%	1%	90%	93%	98%	90%	93%	98%
$n = 50$	1%	2%	4%	95%	96%	98%	97%	96%	98%
$n = 75$	2%	7%	0%	98%	97%	98%	99%	98%	98%
	$\alpha \neq 0$ and $\gamma = 0$								
	H_0^1			H_0^2			H_0^3		
	$T = 50$	$T = 100$	$T = 150$	$T = 50$	$T = 100$	$T = 150$	$T = 50$	$T = 100$	$T = 150$
$n = 25$	92%	99%	99%	1%	3%	4%	94%	95%	99%
$n = 50$	92%	94%	99%	1%	1%	1%	92%	99%	99%
$n = 75$	97%	99%	99%	1%	3%	1%	97%	99%	99%

Table 1: Percentage of times when different spatial hypotheses in Section 5 are rejected using our proposed spatial test under different scenarios. Significance level is set at 5% in all cases.

	$\alpha \neq 0$ and $\gamma \neq 0$			$\alpha = 0$ and $\gamma \neq 0$			$\alpha \neq 0$ and $\gamma = 0$		
	$T = 50$	$T = 100$	$T = 150$	$T = 50$	$T = 100$	$T = 150$	$T = 50$	$T = 100$	$T = 150$
$n = 50$	87%	93%	93%	92%	91%	94%	89%	92%	93%
$n = 100$	89%	94%	94%	92%	94%	95%	89%	94%	94%
$n = 150$	92%	95%	95%	93%	94%	97%	94%	96%	96%

Table 2: Percentage of diagnostic tests in Section 6 that cannot be rejected under different scenarios, when underlying errors are white noise. Significance level is set at 5% in all cases, with $K = 10$ lags considered.

n increases with T fixed. The test performs well in general for different combinations of (n, T) , with power quickly approaches 1 as a increases from 0 to 0.5.

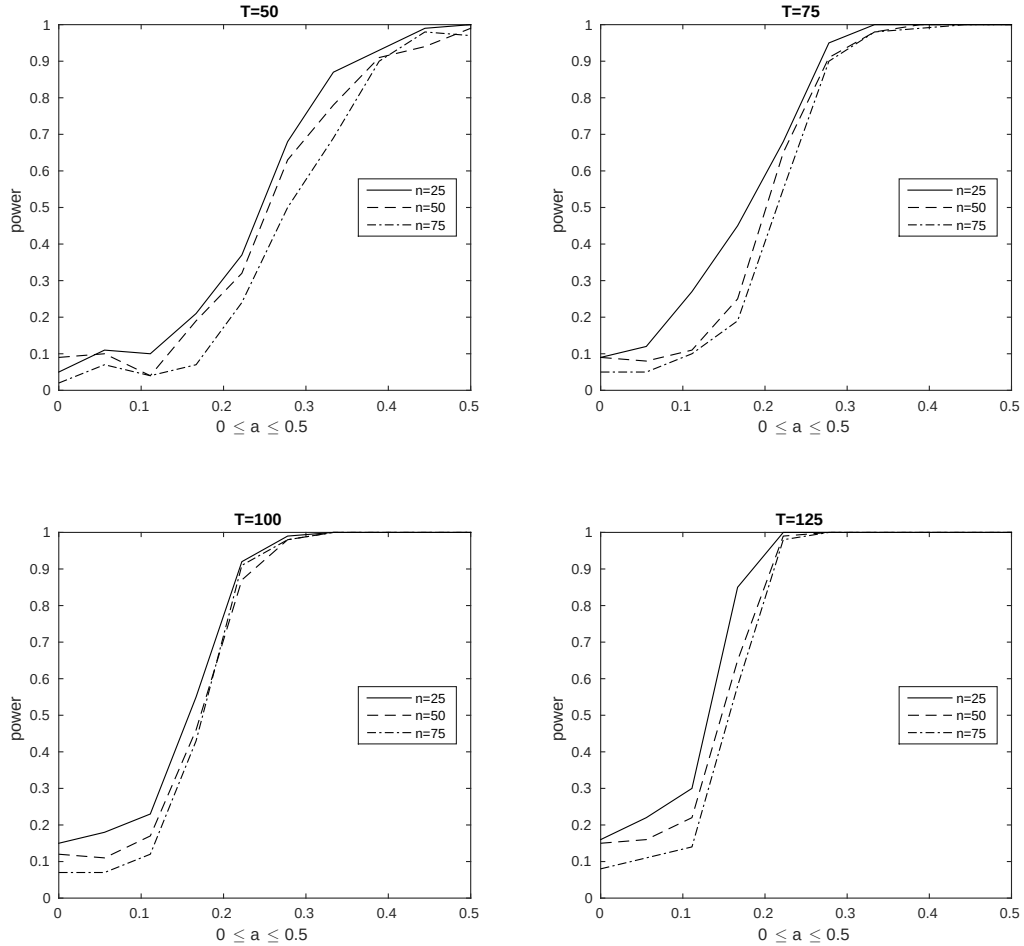


Figure 3: Power curves for the diagnostic test in Section 6 with $0 \leq a \leq 0.5$ for different (n, T) combinations. Significance level is set at 5% in all cases, with $K = 10$ lags considered.

7.4 Stock returns analysis

Although spatial lag models have been used extensively in analyzing economic and geographic data, financial data is rarely analyzed using such models. In fact, stock returns can be envisaged

to be heavily influenced by the sector and country to which its parent company belongs. Indeed, through the use of a spatial lag model with different spatial weights matrix characterizations, Arnold et al. (2013) shows that stock returns belonging to the same country or same industry are spatially related to each other, with inferences on the three spatial correlation parameters in their model. In this section we aim to use our proposed model to analyze the daily log-returns of some stocks in the Euro Stoxx 50 and S&P500, providing inferences to the model parameters, and test if contemporaneous or lag-1 spatial correlations are present, while justifying our final model using our proposed diagnostic test. The stocks we used are listed below.

France	Alstom, Total, BNP, Scociete, Sanofi, Carrefour, LVMH, Vivendi
Germany	Daimler, Allianz, Deutsche Bank
Italy	ENEL, ENI, Intesa, Unicredit, Tele Italy
Spain	Repsol, Banco, Telefonica
US	GM, PG, Nextera, American Express, Citi, Wells Frago, Amgen, Gilead, Johnson, Costco, Home, Centurylink, Verizon
Energy	Alstom, Total, ENEL, ENI, Repsol, PG, Nextera
Finance	BNP, Scociete, Allianz, Deutsche Bank, Intesa, Unicredit, Banco, American Express, Citi, Wells Fargo
Pharmacy	Sanofi, Amgen, Gilead, Johnson
Retails	Carrefour, LVMH, Costco, Home
Telecom	Vivendi, Tele Italy, Telefonica, Centurylink, Verizon
Auto	Daimler, GM

We use the three types of spatial weights matrices used in Arnold et al. (2013) for modeling the contemporaneous as well as the lag-1 spatial correlations in our model setting. The first spatial weights matrix W_1 used has, for $i \neq j$, the (i, j) th entry being the weight in either the Euro Stoxx 50 or the S&P500 index for the j th stock. The second spatial weights matrix W_2 has, for $i \neq j$, the (i, j) th entry being 1 if the two stocks' parent company are in the same country. Finally, the

third spatial weights matrix W_3 is similar to W_2 , except that it is measuring stocks within the same industry branch instead of country. We perform row normalization on all of these three matrices.

We set two covariates, one is the national index, where we use the daily log-return of one of the five national indices S&P500, CAC40, DAX, IBEX or MIB that corresponds to where the parent company of the concerning stock belongs. The other one is the Industry index, where we use the daily log-return of the stock's corresponding industry index in Europe or US. Table 3 shows that

	$\hat{\alpha}_1$	$\hat{\alpha}_2$	$\hat{\alpha}_3$
Value	9.36×10^{-4}	2.2×10^{-3}	1.4×10^{-3}
S.D.	2.9×10^{-7}	4.9×10^{-5}	1.4×10^{-6}
	$\hat{\gamma}_1$	$\hat{\gamma}_2$	$\hat{\gamma}_3$
Value	-1.8×10^{-3}	7.55×10^{-4}	1.8×10^{-3}
S.D.	4.5×10^{-7}	8.1×10^{-7}	5.7×10^{-5}
	$\hat{\phi}$	$\hat{\beta}_1$ (National Index)	$\hat{\beta}_2$ (Industry Index)
Value	1.2×10^{-3}	1.8×10^{-1}	6.9×10^{-1}
S.D.	9.4×10^{-7}	5.4×10^{-7}	4.7×10^{-6}

Table 3: The values of $\hat{\alpha}$, $\hat{\gamma}$, $\hat{\phi}$ and $\hat{\beta}$, with estimated standard deviations (S.D).

all the three spatial weights matrices should be included in our model for both contemporaneous and lag-1 spatial correlations. The contemporaneous spatial correlation is most affected by where the parent company belongs, followed by industry branch, but the reverse is true for the lag-1 spatial correlation. The two covariates are very important too.

Figure 4 shows the heatmaps of the two constructed spatial weights matrices $\sum_{i=1}^3 \hat{\alpha}_i W_i$ and $\sum_{i=1}^3 \hat{\gamma}_i W_i$. For the former, it is obvious that there are some block patterns. These blocks represent the stocks in the same country or in the same industry. Meanwhile, the stocks are related strongly with each other if they are either all from Europe or from US. It is interesting that the ninth stock and twentieth stock are related with each other, although they belong to Germany and US auto industry respectively. The ninth stock is Daimler, while the twentieth stock

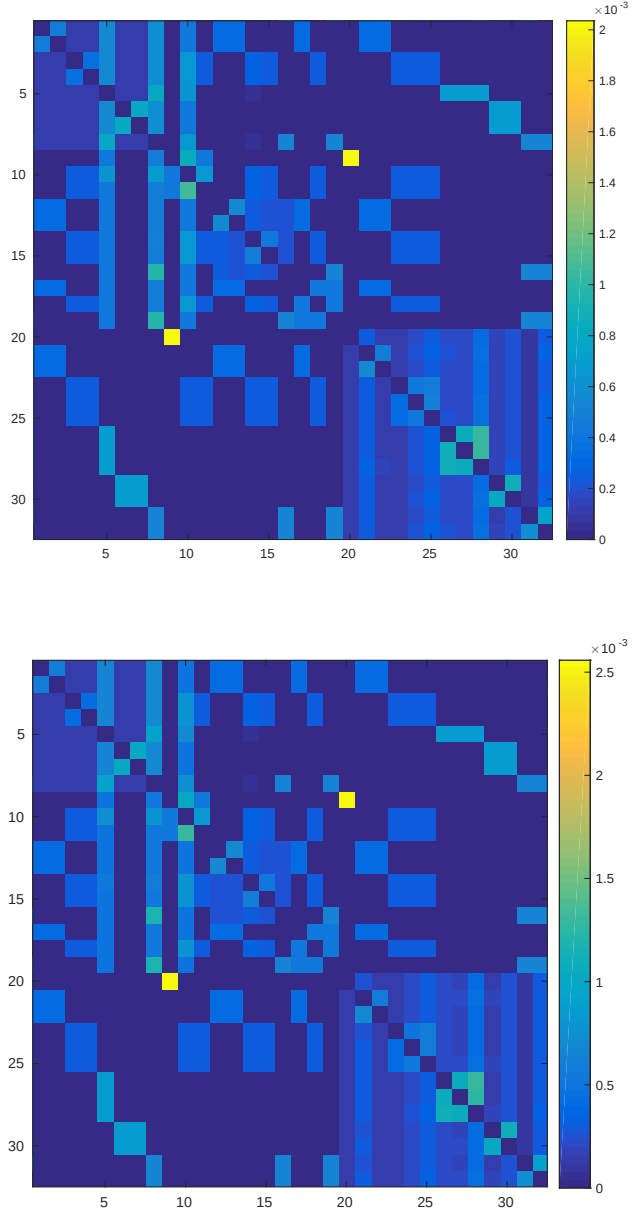


Figure 4: Upper: The matrix $\sum_{i=1}^3 \hat{\alpha}_i \mathbf{W}_i$. Lower: The matrix $\sum_{i=1}^3 \hat{\gamma}_i \mathbf{W}_i$. From 1 to 32, the stocks are Alstom, Total, BNP, Scociete, Sanofi, Carrefour, LVMH, Vivendi, Daimler, Allianz, Deutsche Bank, ENEL, ENI, Intesa, Unicredit, Tele Italy, Repsol, Banco, Telefonica, GM, PG, Nextera, American Express, Citi, Wells Frago, Amgen, Gilead, Johnson, Costco, Home, Centurylink and Verizon respectively.

is GM. But Daimler owns part of GM by spin-offs. This fact means that our linear combination of different spatial weights matrices can reflect a more general pattern and let us learn from it.

Another important feature shown in the first matrix is that there are three bars with larger values comparing with all other areas in this matrix. They represent Sanofi (France Pharmacy), Vivendi (France Telecom) and Allianz (Germany Finance). Therefore, these three stocks have a significant influence on the whole of Europe stock market. The second matrix in Figure 4 also shows a similar pattern.

Finally, we test for spatial correlations in the model and perform diagnostic test. We reject all null hypotheses H_0^1 to H_0^3 in Section 5 due to the approximately zero p -values (4.5×10^{-8} , 2.3×10^{-8} and 1.0×10^{-4} respectively). Therefore, we conclude that these stock returns are under both contemporaneous spatial effects and lag-1 spatial effects. Meanwhile, using the proposed vector white noise test, we cannot reject the null hypothesis that our model residuals are white noise when we consider all lags from 1 to 10.

References

- Abreu, M., de Groot, H. L., and Florax, R. J. (2005). Space And Growth: A Survey Of Empirical Evidence And Methods. *Region et Developpement*, 21:13–44.
- Ahrens, A. and Bhattacharjee, A. (2015). Two-step lasso estimation of the spatial weights matrix. *Econometrics*, 3(1):128.
- Anselin, L. (2010). *Spatial Econometrics: Methods and Models*. Studies in Operational Regional Science. Springer, softcover reprint of hardcover 1st ed. 1988 edition.
- Arnold, M., Stahlberg, S., and Wied, D. (2013). Modeling different kinds of spatial dependence in stock returns. *Empirical Economics*, 44(2):761–774.
- Badinger, H. and Egger, P. (2011). Estimation of higher-order spatial autoregressive cross-section models with heteroskedastic disturbances. *Papers in Regional Science*, 90(1):213–235.
- Baumont, C., Ertur, C., and Le Gallo, J. (2003). *Spatial Convergence Clubs and the European Regional Growth Process, 1980–1995*, pages 131–158. Springer Berlin Heidelberg, Berlin, Heidelberg.

- Bramoullé, Y., Djebbari, H., and Fortin, B. (2009). Identification of peer effects through social networks. *Journal of Econometrics*, 150(1):41 – 55.
- Chang, J., Yao, Q., and Zhou, W. (2017). Testing for vector white noise using maximum cross correlations. *Biometrika*, 104(1):1 – 17.
- Fernandez, V. (2011). Spatial linkages in international financial markets. *Quantitative Finance*, 11(2):237–245.
- Golub, G. H. and van Loan, C. F. (1996). *Matrix Computations*. Johns Hopkins University Press, third edition.
- Horn, R. A. and Johnson, C. R. (2012). *Matrix Analysis*. Cambridge University Press, second edition. Cambridge Books Online.
- Kapoor, M., Kelejian, H. H., and Prucha, I. R. (2007). Panel data models with spatially correlated error components. *Journal of Econometrics*, 140:97–130.
- Lam, C. and Qian, C. (2017). Spatial lag model with time-lagged effects and spatial weight matrix estimation. *Manuscript*.
- Lam, C. and Souza, P. C. L. (2014). Regularization for spatial panel time series using the adaptive lasso. *LSE STICERD, Econometrics Paper Series*.
- Lam, C. and Souza, P. C. L. (2015). Detection and estimation of block structure in spatial weight matrix. *Econometric Reviews*, pages 1–30.
- Lam, C. and Souza, P. C. L. (2016). Spatial lag model estimation with sparse adjustment for spatial weight matrix. *Manuscript*.
- Lee, L. and Liu, X. (2010). Efficient GMM estimation of high order spatial autoregressive models with autoregressive disturbances. *Econometric Theory*, 26:187–230.
- Lee, L. F. and Yu, J. (2010). A spatial dynamic panel data model with both time and individual fixed effects. *Econometric Theory*, 26:564–597.

- Lee, L. F. and Yu, J. (2014). Efficient {GMM} estimation of spatial dynamic panel data models with fixed effects. *Journal of Econometrics*, 180(2):174 – 197.
- LeSage, J. P. and Pace, R. K. (2008). Spatial econometric modeling of origin-destination flows. *Journal of Regional Science*, 48(5):941–967.
- LeSage, J. P. and Pace, R. K. (2009). *Introduction to Spatial Econometrics*. Chapman and Hall.
- Yu, J., de Jong, R., and Lee, L.-f. (2008). Quasi-maximum likelihood estimators for spatial dynamic panel data with fixed effects when both n and t are large. *Journal of Econometrics*, 146(1):118 – 134.
- Zhu, X., Pan, R., Li, G., Liu, Y., and Wang, H. (2016). Network vector autoregression. Unpublished.

8 Technical Proofs

To prove the asymptotic results of the QMLE $\hat{\theta}$, we need to deal with the log likelihood function and its derivatives. Note that the stochastic part of the log likelihood function (4.1) is a function of $\varepsilon_{nt}(\xi)$. For this purpose, we introduce some notations first. Let

$$L_{nt} = (W_{n1}Y_{nt}, W_{n2}Y_{nt}, \dots, W_{nr}Y_{nt}) \quad \text{and} \quad O_{nt} = (L_{n,t-1}, Y_{n,t-1}, X_{nt}). \quad (8.1)$$

According to (2.1) and (4.2), we have

$$\begin{aligned} \varepsilon_{nt}(\xi) &= Y_{nt} - \sum_{i=1}^r \alpha_i W_{ni} Y_{nt} - \sum_{i=1}^r \gamma_i W_{ni} Y_{n,t-1} - \phi Y_{n,t-1} - X_{nt} \beta \\ &= \varepsilon_{nt} - L_{nt}(\alpha - \alpha_0) - O_{nt}(\delta - \delta_0). \end{aligned} \quad (8.2)$$

Due to (4.4), the stochastic part of Y_{nt} has a moving average representation and we define the following two general series to deal with it. Denote

$$\mathcal{U}_{nt} = \sum_{h=1}^{\infty} P_{nh} \varepsilon_{n,t+1-h}, \quad \mathcal{V}_{nt} = \sum_{h=1}^{\infty} Q_{nh} \varepsilon_{n,t+1-h}, \quad (8.3)$$

where P_{nh} and Q_{nh} are sequences of $n \times n$ non-stochastic square matrices. Before giving the proof of Theorem 1, we introduce some useful Lemmas first.

Lemma 1. For $n \times n$ matrices $G_{1n} = (G_{1,ij})$ and $G_{2n} = (G_{2,ij})$, denote $\text{Cov}(\varepsilon'_{nt}G_{1n}\varepsilon_{ns}, \varepsilon'_{ng}G_{2n}\varepsilon_{nh})$ by $\Delta(t, s, g, h)$. Suppose Assumption A2 holds and we have

$$\begin{aligned}\Delta(t, s, g, h) &= (Ee_{11}^4 - 3\sigma_0^4) \sum_{i=1}^n G_{1,ii}G_{2,ii} + \sigma_0^4[\text{tr}(G_{1n}G'_{2n}) + \text{tr}(G_{1n}G_{2n})], \text{ for } t = s = g = h, \\ \Delta(t, s, g, h) &= \Gamma(t - g) \sum_{i=1}^n G_{1,ii}G_{2,ii}, \text{ for } t = s \neq g = h, \\ \Delta(t, s, g, h) &= \Gamma(t - s) \sum_{i=1}^n G_{1,ii}G_{2,ii} + \sigma_0^4\text{tr}(G_{1n}G'_{2n}), \text{ for } t = g \neq s = h, \\ \Delta(t, s, g, h) &= \Gamma(t - s) \sum_{i=1}^n G_{1,ii}G_{2,ii} + \sigma_0^4\text{tr}(G_{1n}G_{2n}), \text{ for } t = h \neq s = g,\end{aligned}$$

and 0 otherwise.

Proof. Under Assumption A2, $\{e_{it}\}$ are uncorrelated across i and t . Thus,

$$E(\varepsilon'_{nt}G_{1n}\varepsilon_{ns}) = \sigma_0^2\text{tr}(G_{1n})I(t = s), \quad (8.4)$$

where $I(t = s) = 1$ for $t = s$ and $I(t = s) = 0$ otherwise.

Note that $\{e_{it}\}$ and $\{e_{jt}\}$ are independent for $i \neq j$ under Assumption A2, and we obtain $E(e_{it}e_{js}e_{pg}e_{qh}) \neq 0$ only if $i = j$ and $p = q$, or $i = p$ and $j = q$, or $i = q$ and $j = p$. Therefore,

$$\begin{aligned}& E\left[\sum_{i,j} e_{it}G_{1,ij}e_{js} \sum_{p,q} e_{pg}G_{1,pq}e_{qh}\right] \\ &= \sum_{i=1}^n G_{1,ii}G_{2,ii}E(e_{it}e_{is}e_{ig}e_{ih}) + \sum_{i=1}^n \sum_{p=1, p \neq i}^n G_{1,ii}G_{2,pp}E(e_{it}e_{is})E(e_{pg}e_{ph}) \\ & \quad + \sum_{i=1}^n \sum_{j=1, j \neq i}^n G_{1,ij}G_{2,ij}E(e_{it}e_{ig})E(e_{js}e_{jh}) + \sum_{i=1}^n \sum_{j=1, j \neq i}^n G_{1,ij}G_{2,ji}E(e_{it}e_{ih})E(e_{js}e_{jg}) \\ &= [E(e_{it}e_{is}e_{ig}e_{ih}) - E(e_{it}e_{is})E(e_{pg}e_{ph}) - E(e_{it}e_{ig})E(e_{js}e_{jh}) - E(e_{it}e_{ih})E(e_{js}e_{jg})] \cdot \\ & \quad \sum_{i=1}^n G_{1,ii}G_{2,ii} + E(e_{it}e_{is})E(e_{pg}e_{ph})\text{tr}(G_{1n})\text{tr}(G_{2n}) \\ & \quad + E(e_{it}e_{ig})E(e_{js}e_{jh})\text{tr}(G_{1n}G'_{2n}) + E(e_{it}e_{ih})E(e_{js}e_{jg})\text{tr}(G_{1n}G'_{2n}).\end{aligned}$$

Combining (8.4), we get for $t = s = g = h$

$$\begin{aligned}\Delta(t, s, g, h) &= E[(\varepsilon'_{nt}G_{1n}\varepsilon_{ns})(\varepsilon'_{ng}G_{2n}\varepsilon_{nh})] - E(\varepsilon'_{nt}G_{1n}\varepsilon_{ns})E(\varepsilon'_{ng}G_{2n}\varepsilon_{nh}) \\ &= (Ee_{11}^4 - 3\sigma_0^2) \sum_{i=1}^n G_{1,ii}G_{2,ii} + \sigma_0^4[\text{tr}(G_{1n}G'_{2n}) + \text{tr}(G_{1n}G_{2n})].\end{aligned}$$

Thus the first equality holds. Noting $E(e_{it}e_{is}) = 0$ for any $t \neq s$, we can similarly prove the other equalities of Lemma 1 hold. $\square$

Lemma 2. *Under Assumptions A2, we have for $t \geq s$*

$$\begin{aligned}
E(\mathcal{U}'_{nt}\mathcal{V}_{ns}) &= \sigma_0^2 \text{tr} \left(\sum_{h=1}^{\infty} P'_{n,t-s+h} Q_{nh} \right) \text{ and} \\
\text{Cov}(\mathcal{U}'_{nt}\mathcal{V}_{nt}, \mathcal{U}'_{ns}\mathcal{V}_{ns}) &= \sum_{h=1}^{t-s} \sum_{g=1}^{\infty} \Gamma(t-s+g-h) \sum_{i=1}^n (P'_{nh} Q_{nh})_{ii} (P'_{ng} Q_{ng})_{ii} \\
&+ \sum_{h=1}^{\infty} \sum_{r=1}^{\infty} \Gamma(h-r) \sum_{i=1}^n (P'_{n,t-s+h} Q_{n,t-s+h})_{ii} (P'_{nr} Q_{nr})_{ii} \\
&+ \sum_{h=1}^{\infty} \sum_{g=1}^{\infty} \left\{ \Gamma(h-g) \sum_{i=1}^n (P'_{n,t-s+h} Q_{n,t-s+g})_{ii} (P'_{nh} Q_{ng})_{ii} + \sigma_0^4 \text{tr} (P'_{n,t-s+h} Q_{n,t-s+g} Q'_{ng} P_{nh}) \right\} \\
&+ \sum_{h=1}^{\infty} \sum_{g=1}^{\infty} \left\{ \Gamma(h-g) \sum_{i=1}^n (P'_{n,t-s+h} Q_{n,t-s+g})_{ii} (P'_{ng} Q_{nh})_{ii} + \sigma_0^4 \text{tr} (P'_{n,t-s+h} Q_{n,t-s+g} Q'_{ng} P_{nh}) \right\} \\
&- \sum_{h=1}^{\infty} \sum_{i=1}^n \left\{ 2\sigma_0^4 (P'_{n,t-s+h} Q_{n,t-s+h})_{ii} (P'_{nh} Q_{nh})_{ii} + \Gamma(0) (P'_{n,t-s+h} Q_{n,t-s+h})_{ii} (P'_{nh} Q_{nh})_{ii} \right. \\
&\left. + \Gamma(0) \sum_{i=1}^n (P'_{n,t-s+h} Q_{n,t-s+h})_{ii} (P'_{nh} Q_{nh})_{ii} \right\}.
\end{aligned}$$

Proof. (i) By (8.4), we have

$$E(\mathcal{U}'_{nt}\mathcal{V}_{ns}) = \sum_{h=1}^{\infty} \sum_{r=1}^{\infty} E(\varepsilon'_{n,t+1-h} P'_{nh} Q_{nr} \varepsilon_{n,s+1-r}) = \sigma_0^2 \text{tr} \left(\sum_{h=1}^{\infty} P'_{n,t-s+h} Q_{nh} \right).$$

(ii) For $t \geq s$, $\mathcal{U}_{nt}$ and $\mathcal{V}_{nt}$ can be rewritten as

$$\mathcal{U}_{nt} = \sum_{h=1}^{t-s} P_{nh} \varepsilon_{n,t+1-h} + \sum_{g=1}^{\infty} P_{n,t-s+g} \varepsilon_{n,s+1-g}, \quad \mathcal{V}_{nt} = \sum_{h=1}^{t-s} Q_{nh} \varepsilon_{n,t+1-h} + \sum_{g=1}^{\infty} Q_{n,t-s+g} \varepsilon_{n,s+1-g}.$$

Under Assumption A2, we have

$$\begin{aligned}
\text{Cov} \left(\sum_{h=1}^{t-s} P_{nh} \varepsilon_{n,t+1-h} \sum_{g=1}^{\infty} Q_{n,t-s+g} \varepsilon_{n,s+1-g}, \mathcal{U}'_{ns} \mathcal{V}_{ns} \right) &= 0, \\
\text{Cov} \left(\sum_{g=1}^{\infty} P_{n,t-s+g} \varepsilon_{n,s+1-g} \sum_{h=1}^{t-s} Q_{nh} \varepsilon_{n,t+1-h}, \mathcal{U}'_{ns} \mathcal{V}_{ns} \right) &= 0.
\end{aligned}$$

Thus,

$$\text{Cov}(\mathcal{U}'_{nt}\mathcal{V}_{nt}, \mathcal{U}'_{ns}\mathcal{V}_{ns}) = \Delta_1 + \Delta_2,$$

where

$$\begin{aligned}\Delta_1 &= Cov\left(\sum_{h=1}^{t-s} \varepsilon'_{n,t+1-h} P'_{nh} \sum_{h=1}^{t-s} Q_{nh} \varepsilon_{n,t+1-h}, \sum_{g=1}^{\infty} \varepsilon'_{n,s+1-g} P'_{ng} \sum_{g=1}^{\infty} Q_{ng} \varepsilon_{n,s+1-g}\right), \\ \Delta_2 &= Cov\left(\sum_{g=1}^{\infty} \varepsilon'_{n,s+1-g} P'_{n,t-s+g} \sum_{g=1}^{\infty} Q_{n,t-s+g} \varepsilon_{n,s+1-g}, \sum_{g=1}^{\infty} P'_{n,g} \varepsilon'_{n,s+1-g} \sum_{g=1}^{\infty} Q_{n,g} \varepsilon_{n,s+1-g}\right).\end{aligned}$$

Using Lemma 1, we obtain

$$\begin{aligned}\Delta_1 &= \sum_{h=1}^{t-s} \sum_{g=1}^{\infty} Cov(\varepsilon'_{n,t+1-h} P'_{nh} Q_{nh} \varepsilon_{n,t+1-h}, \varepsilon'_{n,s+1-g} P'_{ng} Q_{ng} \varepsilon_{n,s+1-g}) \\ &= \sum_{h=1}^{t-s} \sum_{g=1}^{\infty} \Gamma(t-s+g-h) \sum_{i=1}^n (P'_{nh} Q_{nh})_{ii} (P'_{ng} Q_{ng})_{ii}.\end{aligned}$$

Let's turn to Δ_2 then. By Lemma 1, we have

$$\begin{aligned}\Delta_2 &= \sum_{h=1}^{\infty} \sum_{g=1}^{\infty} \sum_{r=1}^{\infty} \sum_{k=1}^{\infty} Cov(\varepsilon'_{n,s+1-h} P'_{n,t-s+h} Q_{n,t-s+g} \varepsilon_{n,s+1-g}, \varepsilon'_{n,s+1-r} P'_{nr} Q_{nk} \varepsilon_{n,s+1-k}) \\ &= \sum_{h=1}^{\infty} Cov(\varepsilon'_{n,t+1-h} P'_{n,t-s+h} Q_{n,t-s+h} \varepsilon_{n,t+1-h}, \varepsilon'_{n,s+1-h} P'_{nh} Q_{nh} \varepsilon_{n,s+1-h}) \\ &\quad + \sum_{h=1}^{\infty} \sum_{r=1, r \neq h}^{\infty} Cov(\varepsilon'_{n,s+1-h} P'_{n,t-s+h} Q_{n,t-s+h} \varepsilon_{n,s+1-h}, \varepsilon'_{n,s+1-r} P'_{nr} Q_{nr} \varepsilon_{n,s+1-r}) \\ &\quad + \sum_{h=1}^{\infty} \sum_{g=1, g \neq h}^{\infty} Cov(\varepsilon'_{n,s+1-h} P'_{n,t-s+h} Q_{n,t-s+g} \varepsilon_{n,s+1-g}, \varepsilon'_{n,s+1-h} P'_{nh} Q_{ng} \varepsilon_{n,s+1-g}) \\ &\quad + \sum_{h=1}^{\infty} \sum_{g=1, r \neq h}^{\infty} Cov(\varepsilon'_{n,s+1-h} P'_{n,t-s+h} Q_{n,t-s+g} \varepsilon_{n,s+1-g}, \varepsilon'_{n,s+1-g} P'_{ng} Q_{nh} \varepsilon_{n,s+1-h}) \\ &=: \Delta_{21} + \Delta_{22} + \Delta_{23} + \Delta_{24},\end{aligned}$$

where

$$\begin{aligned}
\Delta_{21} &= (Ee_{11}^4 - 3\sigma^4) \sum_{h=1}^{\infty} \sum_{i=1}^n (P'_{n,t-s+h} Q_{n,t-s+h})_{ii} (P'_{nh} Q_{nh})_{ii} \\
&\quad + \sum_{h=1}^{\infty} \sigma_0^4 [tr(P'_{n,t-s+h} Q_{n,t-s+h} Q'_{nh} P_{nh}) + tr(P'_{n,t-s+h} Q_{n,t-s+h} P'_{nh} Q_{nh})], \\
\Delta_{22} &= \sum_{h=1}^{\infty} \sum_{r=1}^{\infty} \Gamma(h-r) \sum_{i=1}^n (P'_{n,t-s+h} Q_{n,t-s+h})_{ii} (P'_{nr} Q_{nr})_{ii} \\
&\quad - \Gamma(0) \sum_{h=1}^{\infty} \sum_{i=1}^n (P'_{n,t-s+h} Q_{n,t-s+h})_{ii} (P'_{nh} Q_{nh})_{ii}, \\
\Delta_{23} &= \sum_{h=1}^{\infty} \sum_{g=1}^{\infty} \left\{ \Gamma(h-g) \sum_{i=1}^n (P'_{n,t-s+h} Q_{n,t-s+g})_{ii} (P'_{nh} Q_{ng})_{ii} + \sigma_0^4 tr(P'_{n,t-s+h} Q_{n,t-s+g} Q'_{ng} P_{nh}) \right\} \\
&\quad - \sum_{h=1}^{\infty} \left\{ \Gamma(0) \sum_{i=1}^n (P'_{n,t-s+h} Q_{n,t-s+h})_{ii} (P'_{nh} Q_{nh})_{ii} + \sigma_0^4 tr(P'_{n,t-s+h} Q_{n,t-s+h} Q'_{nh} P_{nh}) \right\} \\
\Delta_{24} &= \sum_{h=1}^{\infty} \sum_{g=1}^{\infty} \left\{ \Gamma(h-g) \sum_{i=1}^n (P'_{n,t-s+h} Q_{n,t-s+g})_{ii} (P'_{ng} Q_{nh})_{ii} + \sigma_0^4 tr(P'_{n,t-s+h} Q_{n,t-s+g} Q'_{ng} P_{nh}) \right\} \\
&\quad - \sum_{h=1}^{\infty} \left\{ \Gamma(0) \sum_{i=1}^n (P'_{n,t-s+h} Q_{n,t-s+h})_{ii} (P'_{nh} Q_{nh})_{ii} + \sigma_0^4 tr(P'_{n,t-s+h} Q_{n,t-s+h} Q'_{nh} P_{nh}) \right\}.
\end{aligned}$$

Therefore, we obtain the results of Lemma 2. $\square$

Lemma 3. Suppose assumptions A1-A7 hold. Then for any $n \times n$ non-stochastic matrix $G_n = (G_{ij})$ with finite L_1 norm and L_{∞} norm, it follows that

$$\begin{aligned}
(i) \quad & \frac{1}{nT} \sum_{t=1}^T \varepsilon'_{nt} G_n \varepsilon_{nt} - E \frac{1}{nT} \sum_{t=1}^T \varepsilon'_{nt} G_n \varepsilon_{nt} = O_p\left(\frac{1}{\sqrt{nT}}\right) \text{ and } E \frac{1}{nT} \sum_{t=1}^T \varepsilon'_{nt} G_n \varepsilon_{nt} = O(1) \\
(ii) \quad & \frac{1}{nT} \sum_{t=1}^T L'_{nt} G_n L_{nt} - E \frac{1}{nT} \sum_{t=1}^T L'_{nt} G_n L_{nt} = O_p\left(\frac{1}{\sqrt{nT}}\right) \text{ and } E \frac{1}{nT} \sum_{t=1}^T L'_{nt} G_n L_{nt} = O(1) \\
(iii) \quad & \frac{1}{nT} \sum_{t=1}^T O'_{nt} G_n O_{nt} - E \frac{1}{nT} \sum_{t=1}^T O'_{nt} G_n O_{nt} = O_p\left(\frac{1}{\sqrt{nT}}\right) \text{ and } E \frac{1}{nT} \sum_{t=1}^T O'_{nt} G_n O_{nt} = O(1) \\
(iv) \quad & \frac{1}{nT} \sum_{t=1}^T L'_{nt} G_n \varepsilon_{nt} - E \frac{1}{nT} \sum_{t=1}^T L'_{nt} G_n \varepsilon_{nt} = O_p\left(\frac{1}{\sqrt{nT}}\right) \text{ and } E \frac{1}{nT} \sum_{t=1}^T L'_{nt} G_n \varepsilon_{nt} = O(1) \\
(v) \quad & \frac{1}{nT} \sum_{t=1}^T O'_{nt} G_n \varepsilon_{nt} = O_p\left(\frac{1}{\sqrt{nT}}\right) \text{ and } E \frac{1}{nT} \sum_{t=1}^T O'_{nt} G_n \varepsilon_{nt} = 0.
\end{aligned}$$

Proof. (i) By Lemma 1, we have

$$\begin{aligned}
\Delta_1 &= \frac{1}{nT} \sum_{t=1}^T \text{var}(\varepsilon'_{nt} G_n \varepsilon_{nt}) \\
&= (Ee_{11}^4 - 3\sigma^2) \frac{1}{n} \sum_{i=1}^n G_{ii}^2 + \sigma^4 \frac{1}{n} [\text{tr}(G_n G'_n) + \text{tr}(G_n^2)] \\
&= O(1)
\end{aligned}$$

due to the infinity of $\|G_n\|_1$ and $\|G_n\|_\infty$. By Assumption A2(ii) and Lemma 1, we have

$$\begin{aligned}
\Delta_2 &= \frac{2}{nT} \sum_{s < t} \text{cov}(\varepsilon'_{nt} G_n \varepsilon_{nt}, \varepsilon'_{ns} G_n \varepsilon_{ns}) \\
&= \frac{2}{nT} \sum_{s < t} \Gamma(t-s) \sum_{i=1}^n G_{ii}^2 \\
&= O(1).
\end{aligned}$$

Hence,

$$E\left(\frac{1}{\sqrt{nT}} \sum_{t=1}^T \varepsilon'_{nt} G_n \varepsilon_{nt} - E \frac{1}{\sqrt{nT}} \sum_{t=1}^T \varepsilon'_{nt} G_n \varepsilon_{nt}\right)^2 = \Delta_1 + \Delta_2 = O(1),$$

which implies the first part of (i) in Lemma 3. Since $\|G_n\|_1$ and $\|G_n\|_\infty$ are both finite, we have

$$E \frac{1}{nT} \sum_{t=1}^T \varepsilon'_{nt} G_n \varepsilon_{nt} = E \frac{\sigma^2}{n} \text{tr}(G_n) = O(1).$$

(ii) Note that the (i, j) element of $L'_{nt} G_n L_{nt}$ is $Y'_{nt} W'_{ni} G_n W_{nj} Y_{nt}$. By (4.4), we get

$$\begin{aligned}
Y'_{nt} W'_{ni} G_n W_{nj} Y_{nt} &= \mathcal{E}'_{nt} W'_{ni} G_n W_{nj} \mathcal{E}_{nt} + \mathcal{X}'_{nt} (W'_{ni} G_n W_{nj} + W'_{nj} G_n W_{ni}) \mathcal{E}_{nt} + \mathcal{X}'_{nt} W'_{ni} G_n W_{nj} \mathcal{X}_{nt} \\
&=: H_{1,nt} + H_{2,nt} + H_{3,nt}.
\end{aligned}$$

To prove the first part of Lemma 3, we only need to prove for $i = 1, 2, 3$,

$$\frac{1}{nT} \sum_{t=1}^T H_{i,nt} - E \frac{1}{nT} \sum_{t=1}^T H_{i,nt} = O\left(\frac{1}{\sqrt{nT}}\right), \quad (8.5)$$

Since $\{\mathcal{X}'_{nt}\}$ are non-stochastic, it is obvious that the left side of (8.5) is zero for $i = 3$. Using lemma 8 of Yu et al. (2008), we can obtain (8.5) holds for $i = 2$.

Next, we deal with the case $i = 1$. By (4.4) and (8.3), we have $H_{1,nt} = \mathcal{U}'_{nt} \mathcal{V}_{nt}$ with $P_{nh} = W_{ni} A_n^h H_n^{-1}$ and $Q_{nh} = G_n W_{nj} A_n^h H_n^{-1}$. Therefore, we only need to prove that

$$E\left\{\frac{1}{\sqrt{nT}} \sum_{t=1}^T H_{i,nt} - E \frac{1}{\sqrt{nT}} \sum_{t=1}^T H_{i,nt}\right\}^2 = \frac{1}{nT} \text{var}\left(\sum_{t=1}^T \mathcal{U}'_{nt} \mathcal{V}_{nt}\right) = O(1). \quad (8.6)$$

On the other hand, by Lemma 2,

$$\begin{aligned} \frac{1}{nT} \text{Var}(\sum_{t=1}^T \mathcal{U}'_{nt} \mathcal{V}_{nt}) &= \frac{1}{nT} \sum_{t=1}^T \sum_{s=1}^T \text{Cov}(\mathcal{U}'_{nt} \mathcal{V}_{nt}, \mathcal{U}'_{ns} \mathcal{V}_{ns}) \\ &=: \frac{1}{nT} \sum_{t=1}^T \sum_{s=1}^T (\mathcal{D}_1(t-s) + \mathcal{D}_2(t-s) + \mathcal{D}_3(t-s) + \mathcal{D}_4(t-s) + \mathcal{D}_5(t-s)), \end{aligned}$$

where $\mathcal{D}_i(t-s)$ is the corresponding part of $\text{Cov}(\mathcal{U}'_{nt} \mathcal{V}_{nt}, \mathcal{U}'_{ns} \mathcal{V}_{ns})$ in Lemma 2. Since $\Gamma(h)$ is summable, under the Assumptions A2-A7, the proof of Lemma 6 of Yu et al. (2008) yields that

$$\frac{1}{nT} \sum_{t=1}^T \sum_{s=1}^T (\mathcal{D}_2(t-s) + \mathcal{D}_3(t-s) + \mathcal{D}_4(t-s) + \mathcal{D}_5(t-s)) = O(1). \quad (8.7)$$

From Lemma 2, we have

$$\begin{aligned} & \left| \frac{1}{nT} \sum_{t=1}^T \sum_{s=1}^T \mathcal{D}_1(t-s) \right| \\ &= \left| \frac{1}{nT} \sum_{t=1}^T \sum_{s=1}^T \sum_{h=1}^{t-s} \sum_{g=1}^{\infty} \Gamma(t-s+g-h) \sum_{i=1}^n (P'_{nh} Q_{nh})_{ii} (P'_{ng} Q_{ng})_{ii} \right| \\ &\leq \frac{1}{n} \sum_{h=1}^{\infty} \sum_{g=1}^{\infty} \sum_{i=1}^n \text{abs}(P'_{nh} Q_{nh})_{ii} \text{abs}(P'_{ng} Q_{ng})_{ii} \frac{1}{n} \sum_{t=1}^T \sum_{s=1}^T |\Gamma(t-s+g-h)| \\ &\leq \frac{1}{n} \sum_{h=1}^{\infty} \sum_{g=1}^{\infty} \sum_{i=1}^n \text{abs}(P'_{nh} Q_{nh})_{ii} \text{abs}(P'_{ng} Q_{ng})_{ii} \frac{1}{T} \sum_{t=1}^T \sum_{j=1}^{\infty} |\Gamma(j)| \\ &= \sum_{j=1}^{\infty} |\Gamma(j)| \frac{1}{n} \sum_{h=1}^{\infty} \sum_{g=1}^{\infty} \sum_{i=1}^n \text{abs}(P'_{nh} Q_{nh})_{ii} \text{abs}(P'_{ng} Q_{ng})_{ii} \\ &= O(1), \end{aligned}$$

due to Assumption A2 and the proof of Lemma 6 of Yu et al. (2008). Combining with (8.7), we have showed that (8.5) holds for $i = 1$ and thus the first part of (ii) of Lemma 3. By Lemma 2 and similar method, we can easily show that the second part of (ii) of Lemma 3 holds.

(iii) Note that X_{nt} is non-stochastic and the same method as (ii) can yield the results.

(iv) By (2.1), we have

$$\varepsilon_{nt} = Y_{nt} - \left(\sum_{i=1}^r \alpha_{0i} W_{ni} \right) Y_{nt} - \left(\sum_{i=1}^r \gamma_{0i} W_{ni} \right) Y_{n,t-1} - \phi_0 Y_{n,t-1} + X_{nt} \beta_0.$$

Using the same method as (ii), we can get the results.

(v) By Assumption A2 and the definition of O_{nt} , we can easily verify

$$E \frac{1}{nT} \sum_{t=1}^T O'_{nt} G_n \varepsilon_{nt} = 0.$$

Using the same method as (ii), we can get the results. $\square$

Lemma 4. Suppose $U_i = (u_{i,ks})$ are $n \times n$ matrices, $i = 1, \dots, r$. It follows that

$$U = \begin{pmatrix} \text{tr}(U_1 U_1') & \text{tr}(U_1 U_2') & \cdots & \text{tr}(U_1 U_r') \\ \text{tr}(U_2 U_1') & \text{tr}(U_2 U_2') & \cdots & \text{tr}(U_2 U_r') \\ \vdots & \vdots & \vdots & \vdots \\ \text{tr}(U_r U_1') & \text{tr}(U_r U_2') & \cdots & \text{tr}(U_r U_r') \end{pmatrix}$$

is non-negative.

Proof. We only need to prove $x' U x \geq 0$ for any r dimensional vector $x = (x_1, \dots, x_r)'$. Note that

$$x' U x = \sum_{i,j} x_i \text{tr}(U_i U_j') x_j = \sum_{i,j} x_i \sum_{k,s} u_{i,ks} u_{j,ks} x_j = \sum_{k,s} \sum_{i,j} u_{i,ks} x_i x_j u_{j,ks} = \sum_{k,s} \left(\sum_i x_i u_{i,ks} \right)^2 \geq 0.$$

Thus, we complete the proof of Lemma 4. $\square$

Lemma 5. Let Θ be any compact parameter space. Then assumptions A1-A7, it follows that

- (i) $l_{nT}(\theta) - El_{nT}(\theta) \xrightarrow{P} 0$ uniformly in Θ ,
- (ii) $El_{nT}(\theta)$ is uniformly equicontinuous for $\theta \in \Theta$.

Proof. (i) By (4.3) and (8.10), we have

$$l_{nT}(\theta) - El_{nT}(\theta) = -\frac{1}{2\sigma^2} \left[\frac{1}{nT} \sum_{t=1}^T \varepsilon'_{nt}(\xi) \varepsilon_{nt}(\xi) - \frac{1}{nT} E \sum_{t=1}^T \varepsilon'_{nt}(\xi) \varepsilon_{nt}(\xi) \right]$$

and Θ is compact implying all parameters are bounded, therefore we only need to prove

$$\frac{1}{nT} \sum_{t=1}^T \varepsilon'_{nt}(\xi) \varepsilon_{nt}(\xi) - \frac{1}{nT} E \sum_{t=1}^T \varepsilon'_{nt}(\xi) \varepsilon_{nt}(\xi) \xrightarrow{P} 0. \quad (8.8)$$

From (2.1) and (4.2), we have

$$\begin{aligned} \varepsilon_{nt}(\xi) &= Y_{nt} - \sum_{i=1}^r \alpha_i W_{ni} - \sum_{i=1}^r \gamma_i W_{ni} Y_{n,t-1} - \phi Y_{n,t-1} - X_{nt} \beta \\ &= L_{nt} \alpha_0 + O_{nt} \delta_0 + \varepsilon_{nt} - L_{nt} \alpha - O_{nt} \delta \\ &= \varepsilon_{nt} - L_{nt} (\alpha - \alpha_0) - O_{nt} (\delta - \delta_0). \end{aligned}$$

Therefore,

$$\begin{aligned}\varepsilon'_{nt}(\xi)\varepsilon_{nt}(\xi) &= \varepsilon'_{nt}\varepsilon_{nt} + (\alpha - \alpha_0)'L'_{nt}L_{nt}(\alpha - \alpha_0) + (\delta - \delta_0)'O'_{nt}O_{nt}(\delta - \delta_0) \\ &\quad - 2(\alpha - \alpha_0)'L'_{nt}O_{nt}(\delta - \delta_0) - 2(\alpha - \alpha_0)'L'_{nt}\varepsilon_{nt} - 2(\delta - \delta_0)'O'_{nt}\varepsilon_{nt}.\end{aligned}\quad (8.9)$$

By Lemma 3, (8.8) holds.

(ii) From (4.3), we have

$$El_{nT}(\theta) = \frac{1}{2} \log 2\pi + \frac{1}{2} \log \sigma^2 - \frac{1}{n} \log |H_n(\alpha)| + \frac{1}{2\sigma^2 nT} E \sum_{t=1}^T \varepsilon'_{nT}(\xi)\varepsilon_{nT}(\xi). \quad (8.10)$$

Since Θ is compact, we have $\frac{1}{2} \log \sigma^2$ is equicontinuous. Therefore, we only need to prove

$$\frac{1}{n} \log |H_n(\alpha)| \text{ is equicontinuous} \quad (8.11)$$

and

$$\frac{1}{2\sigma^2 nT} E \sum_{t=1}^T \varepsilon'_{nT}(\xi)\varepsilon_{nT}(\xi) \text{ is equicontinuous.} \quad (8.12)$$

For any $\alpha^{(1)}$ and $\alpha^{(2)}$, there exists α^* between $\alpha^{(1)}$ and $\alpha^{(2)}$ such that

$$\frac{1}{n} \log |H_n(\alpha^{(1)})| - \frac{1}{n} \log |H_n(\alpha^{(2)})| = \frac{1}{n} \left(\text{tr}(W_{n1}H_n^{-1}(\alpha^*)), \dots, \text{tr}(W_{nr}H_n^{-1}(\alpha^*)) \right) (\alpha^{(2)} - \alpha^{(1)}).$$

By Assumption A5, $\frac{1}{n} \text{tr}(W_{ni}H_n^{-1}(\alpha^*))$ is bounded for $i = 1, \dots, r$, which means (8.11) holds.

From (8.9), Lemma 3 and the compactness of Θ , using the same method as proving (8.11), we can obtain (8.12) holds. $\square$

Lemma 6. *Suppose Assumption A1-A7 hold. If furthermore, e_{it}^2 are uncorrelated across t , then it follows that*

$$\sqrt{nT} \frac{\partial l_{nT}(\theta_0)}{\partial \theta} \xrightarrow{d} N(0, \Omega),$$

where $\Omega = \lim_{T \rightarrow \infty} \Omega_{nT}$ with Ω_{nT} defined by (A.8).

Proof. By (A.8), we know

$$\text{Var}(\sqrt{nT} \frac{\partial l_{nT}(\theta_0)}{\partial \theta}) = \Omega_{nT}.$$

Thus, to prove Lemma 6, it suffices to show that for any nonzero $2r + \kappa_x + 2$ dimensional vector $a = ((a^1)', (a^2)', (a^3)')'$, where a^1 , a^2 and a^3 are r , $r + \kappa_x + 1$ and 1 dimensional vectors,

$$\frac{a' \partial l_{nT}(\theta_0) / \partial \theta}{\sqrt{\text{Var}(a' \partial l_{nT}(\theta_0) / \partial \theta)}} \xrightarrow{d} N(0, 1). \quad (8.13)$$

Let $\bar{B}_n = a_1 B'_{n1} + \dots + a_r B'_{nr}$, we have

$$\begin{aligned} a' nT \frac{\partial l_{nT}(\theta_0)}{\partial \theta} &= -\frac{1}{\sigma_0^2} \sum_{t=1}^T \delta'_0 O'_{nt} \bar{B}'_n \varepsilon_{nt} - \frac{1}{\sigma_0^2} \sum_{t=1}^T [\varepsilon'_{nt} B'_{na} \varepsilon_{nt} - \sigma_0^2 \text{tr}(\bar{B})] \\ &\quad - \frac{1}{\sigma_0^2} \sum_{t=1}^T (a^2)' O'_{nt} \varepsilon_{nt} - \frac{1}{2\sigma_0^2} \sum_{t=1}^T a^3 [\varepsilon'_{nt} \varepsilon_{nt} - n\sigma_0^2] \\ &=: \sum_{t=1}^T \left(\mathcal{U}'_{n,t-1} \varepsilon_{nt} + D'_{nt} \varepsilon_{nt} + \varepsilon'_{nt} M_n \varepsilon_{nt} - \text{tr}(M_n) \right) \\ &= \sum_{t=1}^T \sum_{i=1}^n x_{nt,i}, \end{aligned}$$

where $D_{nt} = \bar{B} X_{nt} \beta_0 + X_{nt} (a_{2r+2}, \dots, a_{2r+\kappa_x+1})$, $M_n = \bar{B} + a_{2r+\kappa_x+2} I_n$, and $\mathcal{U}'_{nt}$ is defined in (8.3) with $P_{nh} = [\bar{B} \Upsilon(\varphi_0) + \Upsilon((a_{r+1}, \dots, a_{2r+\kappa_x+1})')] A_n^h H_n^{-1}$, and

$$x_{nt,i} = (u_{i,t-1} + d_{nti}) e_{it} + m_{n,ii} (e_{it}^2 - \sigma_0^2) + 2 \left(\sum_{j=1}^{i-1} \tilde{m}_{n,ij} e_{jt} \right) e_{it}, \quad (8.14)$$

with $\tilde{m}_{n,ij}$ being the (i, j) th element of $\tilde{M}_n = (M_n + M'_n)/2$. $m_{n,ii}$'s are the diagonal elements of M_n . Since e_{it}^2 are uncorrelated across t , by Assumption A2 and Lemma 1, we obtain that

$$\begin{aligned} \Pi_{nT} &=: \text{Var} \left(a' nT \frac{\partial l_{nT}(\theta_0)}{\partial \theta} \right) \\ &= T \sigma_0^4 \text{tr} \left(\sum_{h=1}^{\infty} P'_{nh} P_{nh} \right) + \sigma_0^2 \sum_{t=1}^T D'_{nt} D_{nt} + T (E e_{11}^4 - 3\sigma_0^4) \sum_{i=1}^n m_{n,ii}^2 \\ &\quad + T \sigma_0^4 (\text{tr}(M_n M'_n) + \text{tr}(M_n^2)) + 2E e_{11}^3 \sum_{t=1}^T \sum_{i=1}^n d_{nti} m_{n,ii}, \end{aligned}$$

where $m_{n,ii}$'s are the diagonal elements of M_n and d_{nti} is the i th element of D_{nt} . Under Assumption A2-A7, using the same method as in Lemma 3, we can verify that

$$\Pi_{nT} = O(nT). \quad (8.15)$$

Define the σ -field

$$\mathcal{F}_{n,t,i} = \sigma(e_{11}, e_{21}, \dots, e_{n1}, \dots, e_{1,t-1}, \dots, e_{n,t-1}, e_{1,t}, \dots, e_{n,t})$$

with $\mathcal{F}_{n,t,0} = \mathcal{F}_{n,t-1,n}$. Let $j = n(t-1) + i$ for $t = 1, \dots, T$ and $i = 1, \dots, n$. Assumption A2 means that $\{x_j\}_{j=1}^{nT}$ form a martingale difference array with respect to $\{\mathcal{F}_{j-1}\}$. According to the CLT for martingale difference in Pansler and Prucha (1997, p.235), it is sufficient to prove that for some

$$\frac{1}{\Pi_{nT}^{1+\tau/4}} \sum_{t=1}^T \sum_{i=1}^n E|x_{nt,i}|^{2+\tau/2} \longrightarrow 0 \text{ and } \frac{1}{\Pi_{nT}} \sum_{t=1}^T \sum_{i=1}^n E(x_{nt,i}^2 | \mathcal{F}_{n,t,i-1}) \longrightarrow 1 \quad (8.16)$$

From (8.14), C_r inequality and Holder inequality together with Assumption A2 imply that

$$\begin{aligned} & E|x_{nt,i}|^{2+\tau/2} \\ & \leq C \left[E|(u_{i,t-1} + d_{nti})e_{it}|^{2+\tau/2} + |m_{n,ii}|^{2+\tau/2} E|(e_{it}^2 - \sigma_0^2)|^{2+\tau/2} + 2E \left| \left(\sum_{j=1}^{i-1} \tilde{m}_{n,ij} e_{jt} \right) e_{it} \right|^{2+\tau/2} \right] \\ & \leq C \left[(E|u_{i,t-1}|^{4+\tau} + |d_{nti}|^{4+\tau}) E|e_{it}|^{4+\tau} \right]^{1/2} + C + 2 \left[E \left| \sum_{j=1}^{i-1} m_{n,ij} e_{jt} \right|^{4+\tau} E|e_{it}|^{4+\tau} \right]^{1/2} \\ & \leq C + C(E|u_{i,t-1}|^{4+\tau})^{1/2} + C \left(\sum_{j=1}^{i-1} |\tilde{m}_{n,ij}| \right)^{1/2}. \end{aligned}$$

Since Assumption A3-A6 ensure the L_1 norm and L_∞ norm of $\sum_{h=1}^\infty P_{nh}$ and M_n are bounded, we have $\sum_{j=1}^{i-1} |m_{n,ij}| \leq C$ and $E|u_{i,t-1}|^{4+\tau} \leq C$ due to Assumption A2. Therefore, $E|x_{nt,i}|^{2+\tau/2}$, which means

$$\sum_{t=1}^T \sum_{i=1}^n E|x_{nt,i}|^{2+\tau/2} = O(nT). \quad (8.17)$$

(8.15) and (8.17) yield

$$\frac{1}{\Pi_{nT}^{1+\tau/4}} \sum_{t=1}^T \sum_{i=1}^n E|x_{nt,i}|^{2+\tau/2} = (nT)^{-\tau/4} \longrightarrow 0.$$

Namely, the first part of (8.16) holds.

On the other hand, by (8.14) we obtain

$$\begin{aligned} E(x_{nt,i}^2 | \mathcal{F}_{n,t,i-1}) &= \sigma_0^2 (u_{n,t-1,i} + d_{nti} + 2 \sum_{j=1}^{i-1} \tilde{m}_{n,ij} e_{jt})^2 + (Ee_{11}^4 - \sigma_0^4) m_{n,ii}^2 \\ &\quad + 2Ee_{11}^3 m_{n,ii} (u_{n,t-1,i} + d_{nti} + 2 \sum_{j=1}^{i-1} \tilde{m}_{n,ij} e_{jt}). \end{aligned}$$

Denote $\tilde{M}_n^-$ be the lower diagonal matrix of $\tilde{M}_n$ and $\tilde{M}_n^o$ be the diagonal matrix with the same diagonal elements as $\tilde{M}_n$, we have

$$\begin{aligned} \sum_{i=1}^n E(x_{nt,i}^2 | \mathcal{F}_{n,t,i-1}) &= \sigma_0^2 (U_{n,t-1} + D_{nt} + \tilde{M}_n^- \varepsilon_{nt})' (U_{n,t-1} + D_{nt} + \tilde{M}_n^- \varepsilon_{nt}) \\ &\quad + (Ee_{11}^4 - \sigma_0^4) \sum_{i=1}^n m_{n,ii}^2 + 2Ee_{11}^3 \tilde{M}_n^o (U_{n,t-1} + D_{nt} + \tilde{M}_n^- \varepsilon_{nt}). \end{aligned}$$

Lemma 3 means the second part of (8.16) holds. Thus, we complete the proof of this lemma. $\square$

Lemma 7. *Under Assumptions A2-A8, $\Sigma = \lim T \rightarrow \infty \Sigma_{nT}$ is nonsingular where Σ_{nT} defined in (A.9).*

Proof. Denote

$$\mathcal{J}_{\alpha\alpha} = E \sum_{t=1}^T \tilde{O}'_{nt} \tilde{O}_{nt}, \quad \mathcal{J}_{\alpha\delta} = E \sum_{t=1}^T \tilde{O}'_{nt} O_{nt}, \quad \mathcal{J}_{\delta\delta} = E \sum_{t=1}^T O'_{nt} O_{nt}.$$

We have

$$E\mathcal{J}_{nT} = \begin{pmatrix} \mathcal{J}_{\alpha\alpha} & \mathcal{J}_{\alpha\delta} \\ \mathcal{J}'_{\alpha\delta} & \mathcal{J}_{\delta\delta} \end{pmatrix}.$$

Assumption A8 means that

$$\mathcal{J}_{\alpha\alpha} - \mathcal{J}_{\alpha\delta} \mathcal{J}_{\delta\delta}^{-1} \mathcal{J}'_{\alpha\delta} \quad \text{is positive definite.} \quad (8.18)$$

Let $a = ((a^1)', (a^2)', (a^3)')'$ be a $2r + \kappa_x + 2$ dimensional vector, where a^1 , a^2 and a^3 are r , $r + \kappa_x + 1$ and 1 dimensional vectors. We only need to prove that $\Sigma a = 0$ implies $a = 0$. By the definition of $\mathcal{J}_{nT}$ and (A.9), we obtain

$$\Sigma = \frac{1}{\sigma_0^2} \begin{pmatrix} \mathcal{J}_{\alpha\alpha} + \lim_{T \rightarrow \infty} \frac{1}{n} (tr(B'_{ni} B_{nj}) + tr(B_{nj} B_{ni}))_{r \times r} & \mathcal{J}_{\alpha\delta} & \lim_{T \rightarrow \infty} \frac{1}{n} \tilde{tr}(B_n) \\ \mathcal{J}'_{\alpha\delta} & \mathcal{J}_{\delta\delta} & 0 \\ \lim_{T \rightarrow \infty} \frac{1}{n} \tilde{tr}(B_n)' & 0 & \frac{1}{2\sigma_0^2} \end{pmatrix}.$$

Therefore, $\Sigma a = 0$ implies

$$\begin{aligned} &\left[\mathcal{J}_{\alpha\alpha} + \lim_{T \rightarrow \infty} \frac{1}{n} (tr(B'_{ni} B_{nj}) + tr(B_{nj} B_{ni}))_{r \times r} \right] a^1 + \mathcal{J}_{\alpha\delta} a^2 + \lim_{T \rightarrow \infty} \frac{1}{n} \tilde{tr}(B_n) a^3 = 0, \\ &a^2 = -\mathcal{J}_{\delta\delta}^{-1} \mathcal{J}'_{\alpha\delta} a^1 \\ &a^3 = -\lim_{T \rightarrow \infty} \frac{2\sigma_0^2}{n} \tilde{tr}'(B_n) a^1. \end{aligned}$$

By eliminating a^2 and a^3 , the above first equation becomes

$$\left\{ \mathcal{J}_{\alpha\alpha} - \mathcal{J}_{\alpha\delta} \mathcal{J}_{\delta\delta}^{-1} \mathcal{J}'_{\alpha\delta} + \lim_{T \rightarrow \infty} \frac{1}{n} \left[\left(\text{tr}(B'_{ni} B_{nj}) + \text{tr}(B_{nj} B_{ni}) \right)_{r \times r} - \frac{2}{n} \tilde{\text{tr}}(B_n) \tilde{\text{tr}}'(B_n) \right] \right\} a^1 = 0.$$

Denote $C_n = (C_{n1}, \dots, C_{nr})'$ with $C_{ni} = B_{ni} - \text{tr}(B_{ni})I_n/n$ and we can verify that

$$\text{tr}(B'_{ni} B_{nj}) + \text{tr}(B_{nj} B_{ni}) - \frac{2}{n} \tilde{\text{tr}}(B_{ni}) \tilde{\text{tr}}(B_{nj}) = \frac{1}{2} \text{tr}[(C_{ni} + C'_{ni})(C_{nj} + C'_{nj})'].$$

By Lemma 4 and (8.18), we know that

$$\mathcal{J}_{\alpha\alpha} - \mathcal{J}_{\alpha\delta} \mathcal{J}_{\delta\delta}^{-1} \mathcal{J}'_{\alpha\delta} + \lim_{T \rightarrow \infty} \frac{1}{n} \left[\left(\text{tr}(B'_{ni} B_{nj}) + \text{tr}(B_{nj} B_{ni}) \right)_{r \times r} - \frac{2}{n} \tilde{\text{tr}}(B_n) \tilde{\text{tr}}'(B_n) \right]$$

is positive definite, which implies that $a^1 = 0$ and hence $a = 0$. Now, we complete the proof of Lemma 7. $\square$

Lemma 8. Suppose Assumptions A2-A8 hold. Then, for any $\theta_* \xrightarrow{P} \theta_0$, we have

$$\frac{\partial^2 l_{nT}(\theta_*)}{\partial \theta \partial \theta'} \xrightarrow{P} \Sigma \quad \text{and} \quad \sqrt{nT} \frac{\partial l_{nT}(\theta_*)}{\partial \theta} \sqrt{nT} \frac{\partial l_{nT}(\theta_*)}{\partial \theta'} \xrightarrow{P} \Omega.$$

Proof. We only prove the first part of this lemma since the second part can be obtained using the same method. Note that

$$\frac{\partial^2 l_{nT}(\theta_*)}{\partial \theta \partial \theta'} - \Sigma = \frac{\partial^2 l_{nT}(\theta_*)}{\partial \theta \partial \theta'} - \frac{\partial^2 l_{nT}(\theta_0)}{\partial \theta \partial \theta'} + \frac{\partial^2 l_{nT}(\theta_0)}{\partial \theta \partial \theta'} - \Sigma_{nT} + \Sigma_{nT} - \Sigma.$$

Since $\Sigma = \lim_{T \rightarrow \infty} \Sigma_{nT}$, it suffices to prove that

$$\frac{\partial^2 l_{nT}(\theta_*)}{\partial \theta \partial \theta'} - \frac{\partial^2 l_{nT}(\theta_0)}{\partial \theta \partial \theta'} \rightarrow 0 \quad \text{and} \quad \frac{\partial^2 l_{nT}(\theta_0)}{\partial \theta \partial \theta'} - \Sigma_{nT} \rightarrow 0. \quad (8.19)$$

From appendix A.2, we have

$$\begin{aligned} \frac{\partial^2 l_{nT}(\theta_*)}{\partial \alpha_i \partial \alpha_j} - \frac{\partial^2 l_{nT}(\theta_0)}{\partial \alpha_i \partial \alpha_j} &= \left(\frac{1}{\sigma_*^2} - \frac{1}{\sigma_0^2} \right) \frac{1}{nT} \sum_{t=1}^T (W_{ni} Y_{nt})' W_{nj} Y_{nt} \\ &\quad + \left[\frac{1}{n\sigma_*^2} \text{Tr}(B_{ni}(\alpha_*) B_{nj}(\alpha_*)) - \frac{1}{n\sigma_0^2} \text{Tr}(B_{ni} B_{nj}) \right] \\ &=: E_1 + E_2. \end{aligned}$$

Since $\sigma_*^2 \rightarrow \sigma_0^2$, Lemma 3 implies $E_1 \xrightarrow{P} 0$. By Assumption A3 and A5, the L_1 and L_∞ norm of $B_{ni}(\alpha_*) B_{nj}(\alpha_*)$ are bounded, which implies that $\frac{1}{n} \text{Tr}(B_{ni}(\alpha_*) B_{nj}(\alpha_*))$ is bounded. On the other

hand,

$$\begin{aligned}
& \frac{1}{n\sigma_0^2} [Tr(B_{ni}(\alpha_*)B_{nj}(\alpha_*)) - Tr(B_{ni}B_{nj})] \\
&= \frac{1}{n\sigma_0^2} Tr[(B_{ni}(\alpha_*) - B_{ni})B_{nj}(\alpha_*) + B_{ni}(B_{nj}(\alpha_*) - B_{nj})] \\
&= \frac{1}{n\sigma_0^2} Tr\left[\sum_{k=1}^r B_{ni}(\tilde{\alpha})B_{nk}(\tilde{\alpha})(\alpha_{*k} - \alpha_{0k})B_{nj}(\alpha_*) + B_{ni}(\alpha_*)\sum_{k=1}^r B_{nj}(\bar{\alpha})B_{nk}(\bar{\alpha})(\alpha_{*k} - \alpha_{0k})\right] \\
&= o_p(1),
\end{aligned}$$

where $\tilde{\alpha}$ and $\bar{\alpha}$ lie between α_* and α_0 , and α_{*k} and α_{0k} are the k th element of α_* and α_0 . Thus,

$$\begin{aligned}
E_2 &= \left(\frac{1}{\sigma_*^2} - \frac{1}{\sigma_0^2}\right) \frac{1}{n} Tr(B_{ni}(\alpha_*)B_{nj}(\alpha_*)) + \frac{1}{n\sigma_0^2} [Tr(B_{ni}(\alpha_*)B_{nj}(\alpha_*)) - Tr(B_{ni}B_{nj})] \\
&= o_p(1)
\end{aligned}$$

Now we have showed that

$$\frac{\partial^2 l_{nT}(\theta_*)}{\partial \alpha \partial \alpha'} - \frac{\partial^2 l_{nT}(\theta_0)}{\partial \alpha \partial \alpha'} \rightarrow 0.$$

Using the same method, we can show the first part of (8.19) holds. Lemma 3 and the expression of $\frac{\partial^2 l_{nT}(\theta_0)}{\partial \alpha \partial \alpha'}$ in appendix imply that the second part of (8.19) holds. Then we complete the proof. $\square$

Proof of Theorem 1: By (2.2) and (4.2), $\varepsilon_{nt}(\xi)$ can be rewritten as

$$\varepsilon_{nt}(\xi) = H_n(\alpha)H_n^{-1}O_{nt}\delta_0 - O_{nt}\delta + H_n(\alpha)H_n^{-1}\varepsilon_{nt}.$$

Due to Lemma 3, we have

$$E\varepsilon'_{nT}(\xi)\varepsilon_{nT}(\xi) = E(H_n(\alpha)H_n^{-1}O_{nt}\delta_0 - O_{nt}\delta)'(H_n(\alpha)H_n^{-1}O_{nt}\delta_0 - O_{nt}\delta) + \sigma_0^2\mathcal{H}_n(\alpha),$$

where $\mathcal{H}_n(\alpha) = tr(H_n^{-1'}H'_n(\alpha)H_n(\alpha)H_n^{-1})$. Therefore, denote $Q_{nt}(\xi) = H_n(\alpha)H_n^{-1}O_{nt}\delta_0 - O_{nt}\delta$ and (8.10) yields

$$El_{nT}(\theta) = \frac{1}{2} \log 2\pi + \frac{1}{2} \log \sigma^2 - \frac{1}{n} \log |H_n(\alpha)| + \frac{\sigma_0^2}{2n\sigma^2} \mathcal{H}_n(\alpha) + \frac{1}{2\sigma^2 nT} \sum_{t=1}^T EQ'_{nt}(\xi)Q_{nt}(\xi).$$

Hence,

$$El_{nT}(\theta) - El_{nT}(\theta_0) = S_{1n} + \frac{1}{2\sigma^2 nT} \sum_{t=1}^T EQ'_{nt}(\xi)Q_{nt}(\xi), \tag{8.20}$$

where $S_{1n} = \frac{1}{2}(\log \sigma^2 - \log \sigma_0^2) - \frac{1}{n}(\log |H_n(\alpha)| - \log |H_n(\alpha_0)|) + \frac{\sigma_0^2}{2n\sigma^2}\mathcal{H}_n(\alpha) - \frac{1}{2}$. Let $p(\alpha, x)$ be the density function of $\tilde{Y}(\alpha, \sigma) = H_n^{-1}(\alpha)\eta_n(\sigma)$, where $\eta_n(\sigma)$ is a multivariate normal variable with mean 0 and covariance $\sigma^2 I_n$. we can easily verify that the Kullback-leibler divergence of $p(\alpha, \sigma)$ from $p(\alpha_0, \sigma_0)$ is S_{1n} , which means $S_{1n} \geq 0$ and the equality holds if and only if $\alpha = \alpha_0$ and $\sigma = \sigma_0$ by the definition of Kullback-leibler divergence. On the other hand, $H_n(\alpha)H_n^{-1} = I_n + \sum_{i=1}^r(\alpha_{0i} - \alpha_i)W_{ni}H_n^{-1}$ means that

$$Q_{nt}(\xi) = O_{nt}(\delta_0 - \delta) + \sum_{i=1}^r(\alpha_{0i} - \alpha_i)W_{ni}H_n^{-1}O_{nt}\delta_0.$$

Since $E\mathcal{J}_{nt}$ is nonsingular, we have $\frac{1}{2\sigma^2 nT} \sum_{t=1}^T EQ'_{nt}(\xi)Q_{nt}(\xi) \geq 0$ and the equality holds if and only if $\xi = \xi_0$. Thus, we have showed that $El_{nT}(\theta)$ has the unique minimum at $\theta = \theta_0$.

Combined with Lemma 5, the consistency of $\hat{\theta}_{nT}$ follows.

Proof of Theorem 2: According to the Taylor expansion, we have

$$\sqrt{nT}(\hat{\theta}_{nT} - \theta_0) = -\left(\frac{1}{nT} \frac{\partial^2 l_{nT}(\bar{\theta}_{nT})}{\partial \theta \partial \theta'}\right)^{-1} \left(\frac{1}{\sqrt{nT}} \frac{\partial l_{nT}(\theta_0)}{\partial \theta}\right).$$

By Lemma 6 and Lemma 8, we obtain the result.

Proof of Theorem 3: Lemma 8 implies that

$$\hat{\Sigma}_{nT} \xrightarrow{p} \Sigma \quad \text{and} \quad \hat{\Omega}_{nT} \xrightarrow{p} \Omega.$$

By Theorem 2, we get the result.

Proof of Theorem 4: According to the proof of Theorem 3 of Chang et al. (2017), we only need to prove

$$\frac{1}{T} \sum_{t=1}^T (\varepsilon_{nt}(\xi) - \varepsilon_{nt})' (\varepsilon_{nt}(\xi) - \varepsilon_{nt}) = o_p(1).$$

By (8.9), Lemma 3 and Theorem 2, we have

$$\begin{aligned}
& \frac{1}{T} \sum_{t=1}^T (\varepsilon_{nt}(\xi) - \varepsilon_{nt})' (\varepsilon_{nt}(\xi) - \varepsilon_{nt}) \\
&= (\alpha - \alpha_0)' \frac{1}{T} \sum_{t=1}^T L'_{nt} L_{nt} (\alpha - \alpha_0) + (\delta - \delta_0)' \frac{1}{T} \sum_{t=1}^T O'_{nt} O_{nt} (\delta - \delta_0) \\
&\quad - 2(\alpha - \alpha_0)' \frac{1}{T} \sum_{t=1}^T L'_{nt} O_{nt} (\delta - \delta_0) \\
&= O_p\left(\frac{1}{nT}\right) O_p(n) \\
&= o_p(1).
\end{aligned}$$

Thus, we complete the proof of Theorem 4.

References

- Abreu, M., de Groot, H. L., and Florax, R. J. (2005). Space And Growth: A Survey Of Empirical Evidence And Methods. *Region et Developpement*, 21:13–44.
- Ahrens, A. and Bhattacharjee, A. (2015). Two-step lasso estimation of the spatial weights matrix. *Econometrics*, 3(1):128.
- Anselin, L. (2010). *Spatial Econometrics: Methods and Models*. Studies in Operational Regional Science. Springer, softcover reprint of hardcover 1st ed. 1988 edition.
- Arnold, M., Stahlberg, S., and Wied, D. (2013). Modeling different kinds of spatial dependence in stock returns. *Empirical Economics*, 44(2):761–774.
- Badinger, H. and Egger, P. (2011). Estimation of higher-order spatial autoregressive cross-section models with heteroskedastic disturbances. *Papers in Regional Science*, 90(1):213–235.
- Baumont, C., Ertur, C., and Le Gallo, J. (2003). *Spatial Convergence Clubs and the European Regional Growth Process, 1980–1995*, pages 131–158. Springer Berlin Heidelberg, Berlin, Heidelberg.
- Bramoullé, Y., Djebbari, H., and Fortin, B. (2009). Identification of peer effects through social networks. *Journal of Econometrics*, 150(1):41 – 55.

- Chang, J., Yao, Q., and Zhou, W. (2017). Testing for vector white noise using maximum cross correlations. *Biometrika*, 104(1):1 – 17.
- Fernandez, V. (2011). Spatial linkages in international financial markets. *Quantitative Finance*, 11(2):237–245.
- Golub, G. H. and van Loan, C. F. (1996). *Matrix Computations*. Johns Hopkins University Press, third edition.
- Horn, R. A. and Johnson, C. R. (2012). *Matrix Analysis*. Cambridge University Press, second edition. Cambridge Books Online.
- Kapoor, M., Kelejian, H. H., and Prucha, I. R. (2007). Panel data models with spatially correlated error components. *Journal of Econometrics*, 140:97–130.
- Lam, C. and Qian, C. (2017). Spatial lag model with time-lagged effects and spatial weight matrix estimation. *Manuscript*.
- Lam, C. and Souza, P. C. L. (2014). Regularization for spatial panel time series using the adaptive lasso. *LSE STICERD, Econometrics Paper Series*.
- Lam, C. and Souza, P. C. L. (2015). Detection and estimation of block structure in spatial weight matrix. *Econometric Reviews*, pages 1–30.
- Lam, C. and Souza, P. C. L. (2016). Spatial lag model estimation with sparse adjustment for spatial weight matrix. *Manuscript*.
- Lee, L. and Liu, X. (2010). Efficient GMM estimation of high order spatial autoregressive models with autoregressive disturbances. *Econometric Theory*, 26:187–230.
- Lee, L. F. and Yu, J. (2010). A spatial dynamic panel data model with both time and individual fixed effects. *Econometric Theory*, 26:564–597.
- Lee, L. F. and Yu, J. (2014). Efficient {GMM} estimation of spatial dynamic panel data models with fixed effects. *Journal of Econometrics*, 180(2):174 – 197.

- LeSage, J. P. and Pace, R. K. (2008). Spatial econometric modeling of origin-destination flows. *Journal of Regional Science*, 48(5):941–967.
- LeSage, J. P. and Pace, R. K. (2009). *Introduction to Spatial Econometrics*. Chapman and Hall.
- Yu, J., de Jong, R., and Lee, L.-f. (2008). Quasi-maximum likelihood estimators for spatial dynamic panel data with fixed effects when both n and t are large. *Journal of Econometrics*, 146(1):118 – 134.
- Zhu, X., Pan, R., Li, G., Liu, Y., and Wang, H. (2016). Network vector autoregression. Unpublished.

A Appendix. Derivatives of the log likelihood function

To deal with the asymptotic distribution of $\hat{\theta}_{NT}$, we give the first-order and second-order derivatives.

A.1 The first order derivatives

From (4.3), we obtain the first order derivatives of $l_{nT}(\theta)$ are

$$\frac{\partial l_{nT}(\theta)}{\partial \alpha} = -\frac{1}{nT\sigma^2} \sum_{t=1}^T [L'_{nt}\varepsilon_{nt}(\xi) - \sigma^2 \tilde{tr}(B_n(\alpha))], \quad (\text{A.1})$$

$$\frac{\partial l_{nT}(\theta)}{\partial \delta} = -\frac{1}{nT\sigma^2} \sum_{t=1}^T O'_{nt}\varepsilon_{nt}(\xi), \quad (\text{A.2})$$

$$\frac{\partial l_{nT}(\theta)}{\partial \sigma^2} = -\frac{1}{2nT\sigma^2} \sum_{t=1}^T [\varepsilon'_{nt}(\xi)\varepsilon_{nt}(\xi) - n\sigma^2]. \quad (\text{A.3})$$

where $\tilde{tr}(B_n(\alpha)) = (tr(B_{n1}(\alpha)), \dots, tr(B_{nr}(\alpha)))'$ and $B_{ni}(\alpha) = W_{ni}H_n^{-1}(\alpha)$, $i = 1, \dots, r$.

Denote $B_{ni} = W_{ni}H_n^{-1}$ and define

$$\tilde{O}_{nt} = (B_{n1}O_{nt}\delta_0, \dots, B_{nr}O_{nt}\delta_0) \quad \text{and} \quad \tilde{tr}(B_n) = (tr(B_{n1}), \dots, tr(B_{nr}))'. \quad (\text{A.4})$$

Note that $Y_{nt} = H_n^{-1}O_{nt}\delta_0 + H_n^{-1}\varepsilon_{nt}$, (A.1)-(A.4) yield

$$\sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \alpha} = -\frac{1}{\sqrt{nT}\sigma_0^2}\sum_{t=1}^T\left\{\tilde{O}'_{nt}\varepsilon_{nt} - [(B_{n1}\varepsilon_{nt}, \dots, B_{nr}\varepsilon_{nt})'\varepsilon_{nt} - \sigma_0^2\tilde{tr}(B_n)]\right\}, \quad (\text{A.5})$$

$$\sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \delta} = -\frac{1}{\sqrt{nT}\sigma_0^2}\sum_{t=1}^TO'_{nt}\varepsilon_{nt}, \quad (\text{A.6})$$

$$\sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \sigma^2} = -\frac{1}{2\sqrt{nT}\sigma_0^2}\sum_{t=1}^T[\varepsilon'_{nt}\varepsilon_{nt} - n\sigma_0^2]. \quad (\text{A.7})$$

Next, we compute the covariance matrix of $\sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \alpha}$. From (A.5)- (A.7)and Assumption A2, we have

$$\begin{aligned} \Omega_{nT,\alpha\alpha} &=: E\sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \alpha} \cdot \sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \alpha'} \\ &= \frac{1}{\sigma_0^2 nT}E\sum_{t=1}^n\tilde{O}'_{nt}\tilde{O}_{nt} + \frac{1}{n}(tr(B'_{ni}B_{nj}) + tr(B'_{ni}B'_{nj}))_{r \times r} \\ &\quad + \frac{Ee_{11}^4 - \sigma_0^4}{n\sigma_0^4}\left(\sum_{k=1}^n B_{ni,kk}B_{nj,kk}\right)_{r \times r}, \\ \Omega_{nT,\alpha\delta} &=: E\sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \alpha} \cdot \sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \delta'} = \frac{1}{\sigma_0^2 nT}E\sum_{i=1}^n\tilde{O}'_{nt}O_{nt}, \\ \Omega_{nT,\alpha\sigma} &=: E\sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \alpha_i} \cdot \sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \sigma^2} = \frac{\tilde{tr}(B_n)}{n} + \frac{Ee_{11}^4 - \sigma_0^4}{2n\sigma_0^4}\tilde{tr}(B_n), \\ \Omega_{nT,\delta\delta} &=: E\sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \delta} \cdot \sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \delta'} = \frac{1}{\sigma_0^2 nT}E\sum_{t=1}^nO'_{nt}O_{nt}, \\ \Omega_{nT,\alpha\sigma} &=: E\sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \delta} \cdot \sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \sigma^2} = 0, \\ \Omega_{nT,\sigma\sigma} &=: E\sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \sigma^2} \cdot \sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \sigma^2} = \frac{1}{2\sigma_0^4} + \frac{Ee_{11}^4 - \sigma_0^4}{4\sigma_0^8}. \end{aligned}$$

Note $E\sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \alpha} = 0$, we have

$$\Omega_{nT} =: Var\left(\sqrt{nT}\frac{\partial l_{nT}(\theta_0)}{\partial \alpha}\right) = \begin{pmatrix} \Omega_{nT,\alpha\alpha} & \Omega_{nT,\alpha\delta} & \Omega_{nT,\alpha\sigma} \\ \Omega'_{nT,\alpha\delta} & \Omega_{nT,\delta\delta} & \Omega_{nT,\delta\sigma} \\ \Omega'_{nT,\alpha\sigma} & \Omega'_{nT,\delta\sigma} & \Omega_{nT,\sigma\sigma} \end{pmatrix}, \quad (\text{A.8})$$

A.2 The second order derivatives

From (A.1)-(A.3), we have

$$\begin{aligned}
\frac{\partial^2 l_{nT}(\theta)}{\partial \alpha \partial \alpha'} &= \frac{1}{nT\sigma^2} \sum_{t=1}^T \left[L'_{nt} L_{nt} + (Tr(B_{ni}(\alpha) B_{nj}(\alpha)))_{r \times r} \right] \\
\frac{\partial^2 l_{nT}(\theta)}{\partial \alpha \partial \delta'} &= \frac{1}{nT\sigma^2} \sum_{t=1}^T L'_{nT} O_{nT} \\
\frac{\partial^2 l_{nT}(\theta)}{\partial \alpha \partial \sigma^2} &= \frac{1}{nT\sigma^4} \sum_{t=1}^T L'_{nt} \varepsilon_{nt}(\xi) \\
\frac{\partial^2 l_{nT}(\theta)}{\partial \delta \partial \delta'} &= \frac{1}{nT\sigma^2} \sum_{t=1}^T O'_{nt} O_{nt} \\
\frac{\partial^2 l_{nT}(\theta)}{\partial \delta \partial \sigma^2} &= \frac{1}{nT\sigma^4} \sum_{t=1}^T O'_{nt} \varepsilon_{nt}(\xi) \\
\frac{\partial^2 l_{nT}(\theta)}{(\partial \sigma^2)^2} &= -\frac{1}{nT\sigma^4} \sum_{t=1}^T \left[-\frac{nT}{2} + \frac{1}{\sigma^2} \sum_{t=1}^T \varepsilon'_{nt}(\xi) \varepsilon_{nt}(\xi) \right]
\end{aligned}$$

Note that $Y_{nt} = H_n^{-1} O_{nt} \delta_0 + H_n^{-1} \varepsilon_{nt}$, by Assumption A2 we have

$$\begin{aligned}
\Sigma_{nT, \alpha\alpha} &=: E \frac{\partial^2 l_{nT}(\theta_0)}{\partial \alpha \partial \alpha'} = \frac{1}{Tn\sigma^2} \sum_{t=1}^T \tilde{O}'_{nt} \tilde{O}_{nt} + \frac{1}{n} (tr(B'_{ni} B_{nj}) + tr(B_{nj} B_{ni}))_{r \times r}, \\
\Sigma_{nT, \alpha\delta} &=: E \frac{\partial^2 l_{nT}(\theta_0)}{\partial \alpha \partial \delta'} = \frac{1}{nT\sigma^2} E \sum_{t=1}^T \tilde{O}'_{nt} O_{nt} \\
\Sigma_{nT, \alpha\sigma} &=: E \frac{\partial^2 l_{nT}(\theta_0)}{\partial \alpha \partial \sigma^2} = \frac{1}{n} tr(B'_{ni}), \\
\Sigma_{nT, \delta\delta} &=: E \frac{\partial^2 l_{nT}(\theta_0)}{\partial \delta \partial \delta'} = \frac{1}{nT\sigma^2} E \sum_{t=1}^T O'_{nt} O_{nt}, \\
\Sigma_{nT, \delta\sigma} &=: E \frac{\partial^2 l_{nT}(\theta_0)}{\partial \delta \partial \sigma^2} = 0, \\
\Sigma_{nT, \sigma\sigma} &=: E \frac{\partial^2 l_{nT}(\theta_0)}{(\partial \sigma^2)^2} = \frac{1}{2\sigma_0^4} - \frac{1}{T\sigma_0^4}.
\end{aligned}$$

Denote the information matrix by Σ_{nT} and we have

$$\Sigma_{nT} =: Var\left(\sqrt{nT} \frac{\partial l_{nT}(\theta_0)}{\partial \alpha}\right) = \begin{pmatrix} \Sigma_{nT, \alpha\alpha} & \Sigma_{nT, \alpha\delta} & \Sigma_{nT, \alpha\sigma} \\ \Sigma'_{nT, \alpha\delta} & \Sigma_{nT, \delta\delta} & \Sigma_{nT, \delta\sigma} \\ \Sigma'_{nT, \alpha\sigma} & \Sigma'_{nT, \delta\sigma} & \Sigma_{nT, \sigma\sigma} \end{pmatrix}. \quad (\text{A.9})$$

Furthermore, comparing with (A.8), we have

$$\Omega_{nT} = \Sigma_{nT} + \Xi_{nT} + O(\frac{1}{T}), \quad (\text{A.10})$$

where

$$\Xi_{nT} = \frac{Ee_{11}^4 - 3\sigma_0^2}{\sigma_0^4} \begin{pmatrix} (\sum_{k=1}^n B_{ni,kk} B_{nj,kk})_{r \times r} & 0_{r \times (r+\kappa_x+1)} & \frac{1}{2n\sigma_0^2} \tilde{tr}(B_n) \\ 0'_{r \times (r+\kappa_x+1)} & 0_{(r+\kappa_x+1) \times (r+\kappa_x+1)} & 0_{(r+\kappa_x+1) \times 1} \\ \frac{1}{2n\sigma_0^2} \tilde{tr}(B_n)' & 0'_{(r+\kappa_x+1) \times 1} & \frac{1}{4\sigma_0^4} \end{pmatrix}.$$